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1 Preface

This new textbook is an additional study material to numerical methods for solution of ordinary differential equations and functional differential equations in subjects Selected Parts from Mathematics (CVPM), Differential Equations in Electrical Engineering (NDRE) and Modern Numerical Methods (NMNM).

The goal of this textbook is to enhance students' knowledge in the area of semi-analytic methods for solution of ordinary differential equations and functional differential equations. Specifically, differential transformation method will be presented. In comparison to other semi-analytic methods like homotopy perturbation method, variational iteration method, Adomian decomposition method and homotopy analysis method, differential transformation method offers simple algebraic recurrent scheme which can be easily implemented without a need for symbolic calculations of derivatives or multidimensional integrals.

This textbook is focused on differential transformation method and its applications in solution of initial value problems for ordinary differential equations, singular initial value problems with nonlinearities, and initial value problems for functional differential equations with proportional and/or constant delay. Moreover, attention is paid to theoretical background of transformation formulas and their use for solution of model exercises.

2 Introduction

A variety of exact, approximate and purely numerical methods are available for solving of ordinary differential equations (ODE), functional differential equations (FDE) and partial differential equations (PDE). Most of these methods are computationally intensive because they are trial-and-error in nature, or need complicated symbolic computations. The differential transformation technique is one of efficient numerical methods for solving ODEs and FDEs. The concept of differential transformation method (DTM) was first proposed by Zhou [15] in 1986 and it was applied to solve linear and nonlinear initial value problems in electric circuit analysis.

This method is a semi-analytical numerical technique that uses Taylor series for solving of differential equations in the form of a polynomial. It is different from high-order Taylor series method which requires symbolic computation of necessary derivatives of data functions. Taylor series method is computationally time-consuming, especially for higher order equations. Differential transformation method is an iterative procedure for obtaining analytic Taylor series solutions of differential equations.

Main advantage of this method is that it can be applied to nonlinear ODEs and FDEs directly, without requiring any linearization, discretization or perturbation. Another important advantage of this method is that it significantly reduces the size of computational work while still providing accurate series solution with fast convergence rate.

3 Differential Transformation Method

3.1 DTM for Ordinary Differential Equations

The differential transformation of the k-th derivative of function $u(t)$ is defined as follows:

$$U(k) = \frac{1}{k!}\left[\frac{d^k u(t)}{dt^k}\right]_{t=t_0}, \tag{3.1.1}$$

where $u(t)$ is the original function and $U(k)$ is the transformed function. Inverse differential transformation of $U(k)$ is defined as follows:

$$u(t) = \sum_{k=0}^{\infty} U(k)(t-t_0)^k, \tag{3.1.2}$$

and hence

$$u(t) = \sum_{k=0}^{\infty} \frac{(t-t_0)^k}{k!}\left[\frac{d^k u(t)}{dt^k}\right]_{t=t_0}. \tag{3.1.3}$$

In fact, inverse transformation (3.1.3) implies that the concept of differential transformation is derived from Taylor series expansion. Although DTM is not able to evaluate the derivatives symbolically, relative derivatives can be calculated by an iterative way which is described by the transformed equations of the original function. Since DTM is based on Taylor series, it is clear that conditions of convergence for DTM are the same as for Taylor series.
In actual applications the function $u(t)$ is expressed as $u_N(t) + R_M$, where a truncated series

$$u_N(t) = \sum_{k=0}^{N} U(k)(t-t_0)^k$$

and R_N is the remainder term given by

$$R_N = \frac{1}{(N+1)!}\frac{d^{N+1}u(t)}{dt^{N+1}}\bigg|_{t=t_1} t^{N+1}$$

for some t_1 such that $0 < t_1 < t$. If the $(N+1)st$ derivative of $u(t)$ is bounded for $t \in (0,1]$, i.e.

$$\left|\frac{d^{N+1}u(t)}{dt^{N+1}}\right| \leq K$$

for a certain nonegative constant K, then the maximum error for $u_N(t)$ in this interval can be evaluated from this reminder term as

$$\mathrm{e}_{max} = \frac{K}{(N+1)!}.$$

From definitions (3.1.1), (3.1.2) we can derive the following formulas for $t_0 = 0$:

Theorem 3.1. *Assume that $F(k)$, $G(k)$, $H(k)$ and $U_i(k)$, $i = 1, \ldots, n$, are differential transformations of functions $f(t)$, $g(t)$, $h(t)$ and $u_i(t)$, $i = 1, \ldots, n$, respectively. Then:*

$$\begin{aligned}
\text{If} f(t) &= \frac{d^n u(t)}{dt^n}, \text{then } F(k) = \frac{(k+n)!}{k!} U(k+n). \\
\text{If} f(t) &= u(t)h(t), \text{then } F(k) = \sum_{l=0}^{k} U(l)H(k-l). \\
\text{If} f(t) &= t^n, \text{then } F(k) = \delta(k-n), \text{ where } \delta \text{ is the Kronecker delta.} \\
\text{If} f(t) &= \mathrm{e}^{\lambda t}, \text{ then } F(k) = \frac{\lambda^k}{k!}. \\
\text{If} f(t) &= \sin\left(at+b\right), \text{then } F(k) = \frac{a^k}{k!} \sin\left(\frac{k\pi}{2} + b\right). \\
\text{If} f(t) &= \cos\left(at+b\right), \text{then } F(k) = \frac{a^k}{k!} \cos\left(\frac{k\pi}{2} + b\right) \\
\text{If} f(t) &= \prod_{i=1}^{n} u_i(t), \text{then} \\
F(k) &= \sum_{s_1=0}^{k} \sum_{s_2=0}^{k-s_1} \cdots \sum_{s_n=0}^{k-s_1-\cdots s_{n-1}} U_1(s_1) \ldots U_{n-1}(s_{n-1}) U_n(k - s_1 - \cdots - s_n).
\end{aligned}$$

Proof. Let us prove for example the first formula:

From definition (3.1.2) we get for $n = 1$

$$u'(t) = \sum_{k=1}^{\infty} kU(k)t^{k-1}.$$

Replacing $k-1$ by k it follows

$$u'(t) = \sum_{k=0}^{\infty} (k+1)U(k+1)t^k.$$

Hence, for $f(t) = g'(t)$, we get

$$F(k) = (k+1)U(k+1).$$

Generally, for $f(t) = \dfrac{d^n u(t)}{dt^n}$, we have

$$\frac{d^n u(t)}{dt^n} = \sum_{k=n}^{\infty} k(k-1)(k-2)\ldots(k-n+1)U(k)t^{k-n}.$$

Replacing $k-n$ by k it follows

$$\frac{d^n u(t)}{dt^n} = \sum_{k=0}^{\infty}(k+n)(k+n-1)(k+n-2)\dots(k+1)U(k+n)t^k = \sum_{k=0}^{\infty}\frac{(k+n)!}{k!}U(k+n)t^k.$$

From here it is obvious that

$$F(k) = \frac{(k+n)!}{k!}U(k+n).$$

The proof is complete.

□

3.2 Numerical Applications for ODEs

Example 1. Consider the following initial value problem

$$u' - u = \mathrm{e}^t, \quad u(0) = 0. \tag{3.2.1}$$

Using formulas i) and iv) from Theorem 3.1 we get

$$(k+1)U(k+1) - U(k) = \frac{1}{k!}, \quad k \geq 0.$$

We obtain the following recurrence relation

$$U(k+1) = \frac{1}{k+1}\left[U(k) + \frac{1}{k!}\right]. \tag{3.2.2}$$

From the initial condition $u(0) = 0$ we have $U(0) = 0$ and from (3.2.2) we get

$$\begin{aligned}
U(1) &= 1, \\
U(2) &= 1, \\
U(3) &= \frac{1}{2} = \frac{1}{2!}, \\
U(4) &= \frac{1}{6} = \frac{1}{3!}, \\
U(5) &= \frac{1}{24} = \frac{1}{4!}, \\
&\vdots
\end{aligned}$$

Hence

$$\begin{aligned}
u(t) &= t + t^2 + \frac{1}{2!}t^3 + \frac{1}{3!}t^4 + \frac{1}{4!}t^5 + \dots \frac{1}{k!}t^{k-1} + \dots \\
&= t\left(1 + t + \frac{1}{2!}t^2 + \frac{1}{3!}t^3 + \frac{1}{4!}t^4 + \dots \frac{1}{k!}t^k + \dots\right).
\end{aligned}$$

In the limit case $k \to \infty$, we can observe that the series solution obtained by differential transformation method converges to the series expansion of the closed form solution

$$u(t) = t\mathrm{e}^t.$$

Example 2. Consider the following initial value problem

$$u'' + u = 0 \tag{3.2.3}$$

subject to initial conditions

$$u(0) = 1, \quad u'(0) = 1. \tag{3.2.4}$$

Taking the differential transformation method of (3.2.3) (see Theorem 3.1), it leads to

$$(k+1)(k+2)U(k+2) + U(k) = 0.$$

Hence

$$U(k+2) = -\frac{U(k)}{(k+1)(k+2)}. \tag{3.2.5}$$

From initial conditions (3.2.4) we have $U(0) = 1,\ U(1) = 1$ and from (3.2.5) we get

$$\begin{aligned}
U(2) &= -\frac{1}{2} = -\frac{1}{2!}, \\
U(3) &= -\frac{1}{6} = -\frac{1}{3!}, \\
U(4) &= \frac{1}{24} = \frac{1}{4!}, \\
U(5) &= \frac{1}{120} = \frac{1}{5!}, \\
U(6) &= -\frac{1}{720} = -\frac{1}{6!}, \\
U(7) &= -\frac{1}{5040} = -\frac{1}{7!}, \\
&\vdots
\end{aligned}$$

Thus

$$\begin{aligned}
u(t) &= 1 + t - \frac{1}{2!}t^2 - \frac{1}{3!}t^3 + \frac{1}{4!}t^4 + \frac{1}{5!}t^5 - \frac{1}{6!}t^6 - \frac{1}{7!}t^7 + \dots \\
&= \left(1 - \frac{1}{2!}t^2 + \frac{1}{4!}t^4 - \frac{1}{6!}t^6 + \dots\right) + \left(t - \frac{1}{3!}t^3 + \frac{1}{5!}t^5 - \frac{1}{7!}t^7 + \dots\right).
\end{aligned}$$

If $k \to \infty$, then the series solution converges to the series expansion of the closed form solution

$$u(t) = \cos t + \sin t.$$

Example 3. Consider the following initial value problem

$$u'' - 3u' + 2u = 2t - 3 \tag{3.2.6}$$

subject to initial conditions

$$u(0) = 1, \quad u'(0) = 2. \tag{3.2.7}$$

Taking the differential transformation method of (3.2.6) we get

$$(k+1)(k+2)U(k+2) - 3(k+1)U(k+1) + 2U(k) = 2\delta(k-2) - \delta(k), \quad k \geq 0. \tag{3.2.8}$$

Thus

$$U(k+2) = \frac{1}{(k+1)(k+2)}\left[3(k+1)U(k+1) - 2U(k) + 2\delta(k-2) - \delta(k)\right].$$

From initial conditions (3.2.7) we have $U(0) = 1,\ U(1) = 2$ and from (3.2.8) we obtain

$$\begin{aligned}
U(2) &= \frac{1}{2} = \frac{1}{2!},\\
U(3) &= \frac{1}{6} = \frac{1}{3!},\\
U(4) &= \frac{1}{24} = \frac{1}{4!},\\
U(5) &= \frac{1}{120} = \frac{1}{5!},\\
&\vdots
\end{aligned}$$

From (3.1.2) we have

$$\begin{aligned}
u(t) &= 1 + 2t + \frac{1}{2!}t^2 + \frac{1}{3!}t^3 + \frac{1}{4!}t^4 + \dots \frac{1}{k!}t^k + \dots\\
&= t + \left(1 + t + \frac{1}{2!}t^2 + \frac{1}{3!}t^3 + \frac{1}{4!}t^4 + \dots \frac{1}{k!}t^k + \dots\right).
\end{aligned}$$

If $k \to \infty$, then the series solution converges to the series expansion of the closed form solution of

$$u(t) = t + \mathrm{e}^t.$$

Example 4. Consider the following initial value problem with variable coefficients

$$u' - 2tu = 0, \quad u(0) = 1. \tag{3.2.9}$$

Using Theorem 3.1 we get

$$(k+1)U(k+1) - 2\sum_{m=0}^{k} \delta(m-1)U(k-m) = 0.$$

Thus

$$U(k+1) = \frac{2}{k+1}\sum_{m=0}^{k} \delta(m-1)U(k-m). \tag{3.2.10}$$

From the initial condition we have $U(0) = 1$. Then for

$$\begin{aligned}
k = 0:\ U(1) &= 2\sum_{m=0}^{0} \delta(m-1)U(-m) = 0 \cdot U(0) = 0.\\
k = 1:\ U(2) &= \sum_{m=0}^{1} \delta(m-1)U(1-m) = [0 \cdot U(1) + 1 \cdot U(0)] = 1.\\
k = 2:\ U(3) &= \frac{2}{3}\sum_{m=0}^{2} \delta(m-1)U(2-m) = [0 \cdot U(2) + 1 \cdot U(1) + 0 \cdot U(0)] = 0.
\end{aligned}$$

$$
\begin{aligned}
k=3:\ U(4) &= \frac{1}{2}\sum_{m=0}^{3}\delta(m-1)U(3-m) = \frac{1}{2}\left[0\cdot U(3)+1\cdot U(2)+0\cdot U(1)+0\cdot U(0)\right] \\
&= \frac{1}{2} = \frac{1}{2!}. \\
k=4:\ U(5) &= \frac{2}{5}\sum_{m=0}^{4}\delta(m-1)U(4-m) \\
&= \left[0\cdot U(4)+1\cdot U(3)+0\cdot U(2)+0\cdot U(1)+0\cdot U(0)\right] = 0. \\
k=5:\ U(6) &= \frac{1}{3}\sum_{m=0}^{5}\delta(m-1)U(5-m) \\
&= \frac{1}{3}\left[0\cdot U(5)+1\cdot U(4)+0\cdot U(3)+0\cdot U(2)+0\cdot U(1)+0\cdot U(0)\right] \\
&= \frac{1}{6} = \frac{1}{3!}.
\end{aligned}
$$

From (3.1.2) we have

$$
\begin{aligned}
u(t) &= 1+t^2+\frac{1}{2!}t^4+\frac{1}{3!}t^6+\frac{1}{4!}t^8+\dots \\
&= 1+t^2+\frac{(t^2)^2}{2!}+\frac{(t^2)^3}{3!}+\frac{(t^2)^4}{4!}+\dots
\end{aligned}
$$

If $k\to\infty$, then the series solution converges to the series expansion of the closed form solution

$$u(t) = \mathrm{e}^{t^2}.$$

3.3 Numerical Applications for Systems of ODE

Example 5. Consider the following system of non-homogeneous differential equations

$$
\begin{aligned}
u_1' &= u_3-\cos t, \\
u_2' &= u_3-\mathrm{e}^t, \\
u_3' &= u_1-u_2,
\end{aligned}
\tag{3.3.1}
$$

subject to initial conditions

$$u_1(0)=1,\ u_2(0)=0,\ u_3(0)=2. \tag{3.3.2}$$

Using Theorem 3.1 system (3.3.1) is transformed as follows:

$$
\begin{aligned}
&(k+1)U_1(k+1)-U_3(k) = -\frac{1}{k!}\cos\left(\frac{k\pi}{2}\right), \\
&(k+1)U_2(k+1)-U_3(k) = -\frac{1}{k!}, \\
&(k+1)U_3(k+1)-U_1(k)+U_2(k) = 0.
\end{aligned}
$$

From initial conditions (3.3.2) we have $U_1(0)=1,\ U_2(0)=0,\ U_3(0)=2.$

Consequently, we find

$$\begin{array}{lll} U_1(1)=1, & U_2(1)=1, & U_3(1)=1, \\ U_1(2)=\frac{1}{2!}, & U_2(2)=0, & U_3(2)=0, \\ U_1(3)=\frac{1}{3!}, & U_2(3)=-\frac{1}{3!}, & U_3(3)=\frac{1}{3!}, \\ U_1(4)=\frac{1}{4!}, & U_2(4)=0, & U_3(4)=\frac{2}{4!}, \\ U_1(5)=\frac{1}{5!}, & U_2(5)=\frac{1}{5!}, & U_3(5)=\frac{1}{5!}, \\ \vdots & & \end{array}$$

From here we have

$$\begin{aligned} u_1(t) &= 1+t+\frac{1}{2!}t^2+\frac{1}{3!}t^3+\frac{1}{4!}t^4+\dots, \\ u_2(t) &= t-\frac{1}{3!}t^3+\frac{1}{5!}t^5-\dots, \\ u_3(t) &= \left(1+t+\frac{1}{2!}t^2+\frac{1}{3!}t^3+\frac{1}{4!}t^4+\dots\right)+\left(1-\frac{1}{2!}t^2+\frac{1}{4!}t^4-\dots\right). \end{aligned}$$

If $k\to\infty$, then the series solution converges to the series expansion of the closed form solution

$$u_1(t)=\mathrm{e}^t, \quad u_2(t)=\sin t, \quad u_3(t)=\mathrm{e}^t+\cos t.$$

Example 6. Consider the following system of non-homogeneous differential equations of the second order

$$u_1''+u_1-u_2''-4u_2=0, \tag{3.3.3}$$
$$u_1'+u_2'=\cos t+2\cos 2t, \tag{3.3.4}$$

subject to initial conditions

$$\begin{aligned} &u_1(0)=0,\ u_1'(0)=1, \\ &u_2(0)=0,\ u_2'(0)=2. \end{aligned} \tag{3.3.5}$$

Applying Theorem 3.1 we get

$$\begin{aligned} &(k+2)(k+1)U_1(k+2)+U_1(k)-(k+2)(k+1)U_2(k+2)-4U_2(k)=0, \\ &(k+1)U_1(k+1)+(k+1)U_2(k+1)=\frac{1}{k!}\cos\left(\frac{k\pi}{2}\right)+\frac{2^{k+1}}{k!}\cos\left(\frac{k\pi}{2}\right). \end{aligned}$$

and from initial conditions

$$\begin{aligned} &U_1(0)=0,\ U_1(1)=1, \\ &U_2(0)=0,\ U_2(1)=2. \end{aligned}$$

Hence

$$\begin{aligned}
&U_1(2) = 0, && U_2(2) = 0,\\
&U_1(3) = -\tfrac{1}{3!}, && U_2(3) = -\tfrac{8}{3!},\\
&U_1(4) = 0, && U_2(4) = 0,\\
&U_1(5) = \tfrac{1}{5!}, && U_2(5) = \tfrac{32}{5!},\\
&U_1(6) = 0, && U_2(6) = 0,\\
&U_1(7) = -\tfrac{1}{7!}, && U_2(7) = \tfrac{128}{5!},\\
&\vdots
\end{aligned}$$

From here we have

$$\begin{aligned}
u_1(t) &= t - \frac{1}{3!}t^3 + \frac{1}{5!}t^5 - \frac{1}{7!}t^7 + \dots,\\
u_2(t) &= 2t - \frac{8}{3!}t^3 + \frac{32}{5!}t^5 - \frac{128}{7!}t^7 + \dots = 2t - \frac{(2t)^3}{3!} + \frac{(2t)^5}{5!} - \frac{(2t)^7}{7!} + \dots
\end{aligned}$$

If $k \to \infty$, then the series solutions converge to the series expansions of the closed form solutions

$$\begin{aligned}
u_1(t) &= \sin t,\\
u_2(t) &= \sin 2t.
\end{aligned}$$

3.4 DTM for Singular Initial Value Problems

Many universal phenomena in mathematical physics and astrophysics which occur on semi-infinite interval can be modelled by the heat equation

$$t^{-k}\frac{d}{dt}\left(t^k\frac{du}{dt}\right) + f(t,u) = h(t), \quad t > 0,\ k > 0, \tag{3.4.1}$$

which can be written in the form

$$u'' + \frac{k}{t}u' + f(t,u) = h(t), \quad t > 0,\ k > 0, \tag{3.4.2}$$

where f, h are nonlinear analytic functions, u is the temperature. Throughout this paper, we assume that u is an analytic function on the given interval.

For the steady-state case, and for $f(t,u) = p(t)q(u),\ k = 2, h(t) = 0$, we obtain the singular initial value problem for generalized Lane-Emden equation

$$u'' + \frac{2}{t}u' + p(t)q(u) = 0, \quad t > 0, \tag{3.4.3}$$

subject to the initial conditions

$$u(0) = a, \quad u'(0) = b, \tag{3.4.4}$$

where $a, b \in \mathbb{R}$. For $p(t) = 1$ we get Lane-Emden type equation which is a model of several phenomena such as temperature variation of a self gravitating star, thermal behaviour of a spherical

cloud of a gas under mutual attraction of its molecules, isothermal process in gas spheres, kinetics of combustion or reactants concentration in chemical reactor (see [1]).

In this section, new formulas for exponential and logarithmic nonlinearities are given which we use for solving certain classes of Lane-Emden type equations describing behavior of stellar structures (see [1],[2]).

Theorem 3.2. *If $f(u(t)) = \mathrm{e}^{au(t)}$, $a \in \mathbb{R}$, then*

$$F(k)) = \begin{cases} \mathrm{e}^{aU(0)}, & k = 0, \\ \frac{a}{k}\sum\limits_{i=1}^{k} iU(i)F(k-i), & k \geq 1, \end{cases} \tag{3.4.5}$$

where

$$U(i) = \frac{1}{i!}\left[\frac{d^i u(t)}{dt^i}\right]_{t=0}, \ i = 1, \ldots k.$$

Proof. From the definition of transform

$$F(0) = \mathrm{e}^{au(t)}\Big|_{t=0} = \mathrm{e}^{au(0)} = \mathrm{e}^{aU(0)}.$$

Put

$$f(u(t)) = \mathrm{e}^{au(t)} = \sum_{k=0}^{\infty} F(k)t^k, \quad u(t) = \sum_{k=0}^{\infty} U(k)t^k. \tag{3.4.6}$$

By differentiation $f(u(t))$ with respect t, we get

$$(f(u(t))' = au'(t)f(u(t)). \tag{3.4.7}$$

From (3.4.6) and Theorem 3.1, we obtain

$$\sum_{k=0}^{\infty}(k+1)F(k+1)t^k = a\left(\sum_{k=0}^{\infty}(k+1)U(k+1)t^k\right)\left(\sum_{k=0}^{\infty}F(k)t^k\right).$$

Comparing the terms with the same power of t^k, we have

$$(k+1)F(k+1) = a\sum_{i=0}^{k}(i+1)U(i+1)F(k-i).$$

From here we get

$$F(k+1) = \frac{a}{k+1}\sum_{i=0}^{k}(i+1)U(i+1)F(k-i), \quad k \geq 0. \tag{3.4.8}$$

Replacing $k+1$ by k and $i+1$ by i it follows

$$F(k) = \frac{a}{k}\sum_{i=1}^{k} iU(i)F(k-i), \quad k \geq 1.$$

The proof is complete.

□

Example 7. Consider the isothermal gas spheres equation which is described by Lane-Emden type equation

$$u'' + \frac{2}{t}u' + \mathrm{e}^u = 0 \tag{3.4.9}$$

with initial conditions $u(0) = u'(0) = 0$.

From initial conditions and Theorem 3.1, we get

$$U(0) = 0,\ U(1) = 0,\ F(0) = 1,\ F(1) = 0.$$

Multiplying equation (3.4.9) by t and using Theorem 3.1, we obtain

$$\begin{aligned}&\sum_{l=0}^{k} \delta(l-1)(k+2-l)(k+1-l)U(k+2-l) + 2(k+1)U(k+1)\\ &+ \sum_{l=0}^{k} \delta(l-1)F(k-l) = 0.\end{aligned} \tag{3.4.10}$$

From here we obtain the recurrence relation

$$U(k+1) = -\frac{1}{(k+1)(k+2)}F(k-1), \quad k \geq 1. \tag{3.4.11}$$

Replacing $k-1$ by k we get

$$U(k+2) = -\frac{1}{(k+2)(k+3)}F(k), \quad k \geq 0. \tag{3.4.12}$$

Using (3.4.5) we can write recurrence relation (3.4.12) in the form

$$U(k+2) = -\frac{1}{k(k+2)(k+3)}\sum_{i=1}^{k} iU(i)F(k-i), \quad k \geq 0. \tag{3.4.13}$$

Since

$$F(k) = -(k+2)(k+3)U(k+2)$$

the relation (3.4.13) can be written in the form

$$U(k+2) = \frac{1}{k(k+2)(k+3)}\sum_{i=1}^{k} i(k-i+2)(k-i+3)U(i)U(k-i+2), \quad k \geq 1. \tag{3.4.14}$$

It is obvious that

$$U(2) = -\frac{1}{6}F(0) = -\frac{1}{6}.$$

Then recurrence relation (3.4.14) gives

$$\begin{aligned}U(3) &= \frac{1}{12}(6U(1)U(2)) = 0,\\ U(4) &= \frac{1}{40}(12U(1)U(3) + 12U(2)U(2)) = \frac{1}{120},\\ U(5) &= \frac{1}{90}(20U(1)U(4) + 24U(2)U(3) + 18U(3)U(2)) = 0,\end{aligned}$$

$$U(6) = \frac{1}{168}\left(30U(1)U(5) + 40U(2)U(4) + 36U(3)^2 + 24U(4)U(2)\right) = -\frac{1}{1890},$$
$$\vdots$$

Using the inverse transformation rule, we obtain approximate solution of equation (3.4.9) in the form

$$u(t) = \sum_{k=0}^{\infty} U(k)t^k = -\frac{1}{6}t^2 + \frac{1}{120}t^4 - \frac{1}{1890}t^6 + \dots$$

Example 8. Now we consider more general type of equation (3.4.9)

$$u'' + \frac{2}{t}u' + 4(2\mathrm{e}^u + \mathrm{e}^{\frac{u}{2}}) = 0 \tag{3.4.15}$$

with initial conditions $u(0) = u'(0) = 0$.

Put

$$\mathrm{e}^u = \sum_{k=0}^{\infty} F_1(k)t^k, \qquad \mathrm{e}^{\frac{u}{2}} = \sum_{k=0}^{\infty} F_2(k)t^k. \tag{3.4.16}$$

Multiplying equation (3.4.15) by t and using Theorem 3.1. we obtain the recurrence relation

$$U(k+1) = -\frac{1}{(k+1)(k+2)}\Big(8F_1(k-1) + 4F_2(k-1)\Big), \quad k \geq 1. \tag{3.4.17}$$

Replacing $k-1$ by k we get

$$U(k+2) = -\frac{1}{(k+2)(k+3)}\Big(8F_1(k) + 4F_2(k)\Big), \quad k \geq 0. \tag{3.4.18}$$

In cases of a linear combination of several nonlinearities it is better to solve such type of equations the following way:

From initial conditions and Theorem 3.2, we have

$$U(0) = U(1) = 0, \; F_1(0) = F_2(0) = 1.$$

Relation (3.4.18) yields

$$U(2) = -\frac{1}{6}(8F_1(0) + 4F_2(0)) = -2.$$

Then

$$F_1(1) = U(1)F_1(0) = 0, \quad F_2(1) = \frac{1}{2}U(1)F_2(0) = 0, \quad U(3) = 0.$$

Following the same procedure, $U(k)$, $k \geq 4$, can be solved as follows:

$$\begin{aligned}
&F_1(2) = -2, \quad F_2(2) = -1, \quad U(4) = 1,\\
&F_1(3) = 0, \quad F_2(3) = 0, \quad U(5) = 0,\\
&F_1(4) = 3, \quad F_2(4) = 1, \quad U(6) = \tfrac{2}{3}.\\
&\vdots
\end{aligned}$$

Using the inverse transformation rule, we obtain the solution of equation (3.4.15) in the form

$$u(t) = -2t^2 + t^4 - \frac{2}{3}t^6 + \dots.$$

In the limit case $k \to \infty$, we can observe that the series solution obtained by differential transformation method converges to the series expansion of the closed form solution

$$u(t)) = -2\ln(1+t^2).$$

Now, we derive the differential transformation of the certain logarithmic nonlinearity of Lane-Emden type equation which occur in the stellar structure theory and the thermionic current theory.

Theorem 3.3. *If $g(u(t)) = u(t)\ln u(t)$, then*

$$\begin{aligned} G(0) &= 0, \\ G(k+1) &= \frac{1}{k+1}\sum_{i=0}^{k}(k-i+1)U(i)U(k-i+1) \\ &+ (k-2i+1)G(i)U(k-i+1), \quad k \geq 1. \end{aligned} \tag{3.4.19}$$

Proof. Let

$$u(t)\ln u(t) = \sum_{k=0}^{\infty} G(k)t^k, \quad u(t) = \sum_{k=0}^{\infty} U(k)t^k. \tag{3.4.20}$$

It is obvious that $G(0) = 0$. Now we use the following identity

$$(ug(u(t)))' = (u + g(u(t)))u'. \tag{3.4.21}$$

From (3.4.20) and Theorem 3.3, we obtain

$$\left(\sum_{k=0}^{\infty} U(k)t^k\right)\left(\sum_{k=0}^{\infty}(k+1)G(k+1)\right) = \left(\sum_{k=0}^{\infty}(U(k)+G(k))t^k\right)\left(\sum_{k=0}^{\infty}(k+1)U(k+1)t^k\right).$$

Comparing the terms with the same power of t^k, we have

$$\sum_{i=0}^{k}(i+1)G(i+1)U(k-i) = \sum_{i=0}^{k}(G(i)+U(i))U(k-i+1)(k-i+1). \tag{3.4.22}$$

From (3.4.22) we get

$$(k+1)G(k+1) = \sum_{i=0}^{k}(G(i)+U(i))U(k-i+1)(k-i+1) - \sum_{i=0}^{k-1}(i+1)G(i+1)U(k-i). \tag{3.4.23}$$

Replacing $i+1$ by i in the second sum on the right-hand side of identity (3.4.23) and the fact that $G(0) = 0$ we obtain

$$G(k+1) = \frac{1}{k+1}\sum_{i=0}^{k}(k-i+1)U(i)U(k-i+1) + (k-2i+1)G(i)U(k-i+1)).$$

The proof is complete.

□

Example 9. Consider the following Lane-Emden type equation

$$u'' + \frac{8}{t}u' + 18u + 4u\ln u = 0 \tag{3.4.24}$$

with initial conditions $u(0) = 1,\ u'(0) = 0$.

Multiplying equation (3.4.24) by t and using Theorem 3.1, we obtain

$$\begin{aligned}&\sum_{l=0}^{k}\delta(l-1)(k+2-l)(k+1-l)U(k+2-l) + 8(k+1)U(k+1)\\&+18\sum_{l=0}^{k}\delta(l-1)U(k-l) + \sum_{l=0}^{k}\delta(l-1)G(k-l) = 0.\end{aligned} \tag{3.4.25}$$

Thus

$$k(k+1)U(k+1) + 8(k+1)U(k+1) + 18U(k-1) + 4G(k-1) = 0.$$

From here it follows

$$U(k+1) = -\frac{1}{(k+1)(k+8)}\left(18U(k-1) + 4G(k-1)\right), \quad k \geq 1. \tag{3.4.26}$$

Replacing $k-1$ by k relation (3.4.26) gives

$$U(k+2) = -\frac{1}{(k+2)(k+9)}\left(18U(k) + 4G(k)\right), \quad k \geq 0, \tag{3.4.27}$$

and from here we get

$$G(k) = -\frac{1}{4}\left((k+2)(k+9)U(k+2) + 18U(k)\right), \quad k \geq 0. \tag{3.4.28}$$

Using (3.4.19) for $k \geq 1$ we can write relation (3.4.27) in the form

$$\begin{aligned}U(k+2) &= -\frac{1}{(k+2)(k+9)}\\&\times\left[18U(k) + \frac{4}{k}\sum_{i=0}^{k-1}\Big((k-i)(U(i)U(k-i) + (k-2i)G(i)U(k-i)\Big)\right],\end{aligned}$$

which in view of (3.4.28) implies

$$\begin{aligned}U(k+2) &= -\frac{1}{(k+2)(k+9)}\left[18U(k) + \frac{4}{k}\sum_{i=0}^{k-1}\Big((k-i)(U(i)U(k-i)\right.\\&-\left.\frac{1}{4}\Big(U(i+2)(i+2)(i+9) + 18U(i)\Big)(k-2i)U(k-i)\Big)\right].\end{aligned} \tag{3.4.29}$$

From initial conditions and relation (3.4.27) we get

$$U(0) = 1,\ U(1) = 0,\ U(2) = -\frac{1}{18}(18U(0) + 4G(0) = -1.$$

Then recurrence relation (3.4.29) yields

$$\begin{aligned} U(3) &= -\frac{1}{30}[18U(1) + +4\Big(U(0)U(1) - \frac{1}{4}\Big(U(2)18 + 18U(0)\Big)U(1)\Big)] = 0, \\ U(4) &= -\frac{1}{44}\Big[18U(2) + 2\Big(2U(0)U(2) - 1/4\Big(U(2)18 + 18U(0)\Big)2U(2)\Big) \\ &+ 2\Big(U(1)^2 - \frac{1}{4}\Big(U(3)30 + 18U(1))\Big)U(1)\Big)\Big] = \frac{1}{2}. \end{aligned}$$

Continuing in this way we obtain

$$U(5) = 0,\ U(6) = -13!,\ U(7) = 0,\ U(8) = \frac{1}{4!}, \ldots$$

Hence the solution of equation has the following form

$$u(t) = 1 - t^2 + \frac{t^4}{2!} - \frac{t^6}{3!} + \frac{t^8}{4!} - \ldots$$

If $k \to \infty$, then the series solution converges to the series expansion of the closed form solution

$$u(t) = \mathrm{e}^{-t^2}.$$

3.5 General Formula for DTM with Nonlinearities

In this section, we present efficient computational approach for solving nonlinear ordinary differential equations with any nonlinearities based on DTM and the modified general formula for Adomian polynomials with differential transformation components. To illustrate the capability and efficiency of the method, examples for certain classes of nonlinear ordinary differential equations including singular initial value problems are solved.

Firstly, we shortly recall Adomian decomposition method (see [3],[4],[5]) for understanding a construction of the general formula for differential transformation components which is based on so-called Adomian polynomials.

Consider the operator equation

$$Lu + Ru + Nu = g \tag{3.5.1}$$

with prescribed auxiliary conditions, where u is the unknown function, L is the highest order derivative which is assumed to be easily invertible, R is a linear differential operator of order less than L, Nu represents the nonlinear terms, and g is the source term. Applying the inverse operator L^{-1} to both sides of equation (3.5.1) and using the given conditions we obtain

$$u = v - L^{-1}(Ru) - L^{-1}(Nu)$$

where the function v represents the terms arising from integrating the source term g and from using the auxiliary conditions. Standard Adomian method defines the solution u by an infinite series

$$u = \sum_{n=0}^{\infty} u_n.$$

The nonlinear term Nu is usually expressed as a nonlinear function $f(u)$. Generally non-linear function $f(u)$, assumed to be an analytic function, is decomposed as follows:

$$f(u) = \sum_{n=0}^{\infty} A_n(u_0, u_1 \dots u_n),$$

where A_n are appropriate Adomian polynomials which are generated according to algorithm determined, for example, in [3],[4],[5]. Adomian polynomials A_n depend, of course, on the particular non-linearity and are calculated by general formula

$$A_n(u_0, u_1 \dots u_n) = \frac{1}{n!}\frac{d^n}{d\lambda^n}\left[f\left(\sum_{k=0}^{\infty}\lambda^k u_k\right)\right]_{\lambda=0}, \quad n \geq 0. \tag{3.5.2}$$

Hence

$$\begin{aligned} A_0 &= f(u_0), \\ A_1 &= u_1 f'(u_0), \\ A_2 &= u_2 f'(u_0) + \frac{u_1^2}{2!} f''(u_0), \\ A_3 &= u_3 f'(u_0) + u_1 u_2 f''(u_0) + \frac{u_1^3}{3!} f'''(u_0), \\ &\vdots \end{aligned} \tag{3.5.3}$$

The components $u_0, u_1, u_2, \dots,$ are thus determined as follows:

$$\begin{aligned} u_0 &= v, \\ u_{k+1} &= -L^{-1}Ru_k - L^{-1}A_k, \ k \geq 0, \end{aligned}$$

where u_0 is referred to as the zeroth component.

Now, we show that the differential transformation components for nonlinear terms are similar to the Adomian polynomials in which each solution component u_i is replaced by the corresponding differential transformations component $U(i), \ i = 0, 1, 2, \dots$

According to the differential transformation, the nonlinear function $f(u)$, which represents nonlinear terms, has the form

$$f(u) = \sum_{k=0}^{\infty} F(k)t^k, \tag{3.5.4}$$

$$u = \sum_{k=0}^{\infty} U(k)t^k. \tag{3.5.5}$$

Substituting (3.5.5) into (3.5.4), we get

$$f\left(\sum_{k=0}^{\infty} U(k)t^k\right) = \sum_{k=0}^{\infty} F(k)t^k. \tag{3.5.6}$$

Expanding the left-hand side of equation (3.5.6) in Taylor series at the point $t = 0$, we obtain

$$f\left(\sum_{k=0}^{\infty} U(k)t^k\right) = f\left(\sum_{k=0}^{\infty} U(k)t^k\right)\Bigg|_{t=0} + t\left[f'\left(\sum_{k=0}^{\infty} U(k)t^k\right)\left(\sum_{k=1}^{\infty} kU(k)t^{k-1}\right)\right]_{t=0}$$

$$
\begin{aligned}
&+ \frac{t^2}{2!}\left[f''\left(\sum_{k=0}^{\infty}U(k)t^k\right)\left(\sum_{k=0}^{\infty}kU(k)t^{k-1}\right)^2\right.\\
&+ \left. f'\left(\sum_{k=0}^{\infty}U(k)t^k\right)\left(\sum_{k=2}^{\infty}U(k)k(k-1)t^{k-2}\right)\right]_{t=0}\\
&+ \frac{t^3}{3!}\left[f'''\left(\sum_{k=0}^{\infty}U(k)t^k\right)\left(\sum_{k=0}^{\infty}kU(k)t^{k-1}\right)^3\right.\\
&+ 3f''\left(\sum_{k=0}^{\infty}U(k)t^k\right)\left(\sum_{k=1}^{\infty}kU(k)t^{k-1}\right)\left(\sum_{k=2}^{\infty}k(k-1)U(k)t^{k-2}\right)\\
&+ \left. f'\left(\sum_{k=0}^{\infty}U(k)t^k\right)\left(\sum_{k=3}^{\infty}k(k-1(k-2)U(k)t^{k-3}\right)\right]_{t=0} + \dots\\
&= f(U(0)) + U(1)f'(U(0))t + [U(2)f'(U(0)) + \frac{1}{2!}U^2(1)f''(U(0))]t^2\\
&+ [U(3)f'(U(0)) + U(1)U(2)f''(U(0)) + \frac{1}{3!}U^3(1)f'''(U(0))]t^3 + \dots\\
&= \sum_{k=0}^{\infty}F(k)t^k.
\end{aligned}
$$

Comparing coefficients with the same powers of t, we get the following differential transformation components

$$
\begin{aligned}
F(0) &= f(U(0)) = A_0(U(0)),\\
F(1) &= U(1)f'(U(0)) = \frac{d}{dt}\left[f\left(\sum_{i=0}^{\infty}U(k)t^i\right)\right]_{t=0} = A_1(U(0),U(1)),\\
F(2) &= U(2)f'(U(0)) + \frac{1}{2!}U^2(1)f''(U(0)) = \frac{1}{2!}\frac{d^2}{dt^2}\left[f\left(\sum_{i=0}^{\infty}U(k)t^i\right)\right]_{t=0}\\
&= A_2(U(0),U(1),U(2)),\\
F(3) &= U(3)f'(U(0)) + U(1)U(2)f''(U(0)) + \frac{1}{3!}U^3(1)f'''(U(0))\\
&= \frac{1}{3!}\frac{d^3}{dt^3}\left[f\left(\sum_{i=0}^{\infty}U(k)t^i\right)\right]_{t=0} = A_3(U(0),U(1),U(2),U(3)),\\
&\vdots
\end{aligned}
\tag{3.5.7}
$$

From recurrence equations (3.5.7), it is obvious that the differential transformation components $F(k),\ k \geq 0$ are similar to the Adomian polynomials (3.5.3) where solution components $u_i(x)$ are replaced by the components $U(i),\ i \geq 0$. Hence, each component $F(k)$ can be obtained by differentiating both sides of equation (3.5.6) k times with respect to t and then setting $t = 0$, which gives the following general formula:

$$
F(k) = \frac{1}{k!}\frac{d^k}{dt^k}\left[f\left(\sum_{i=0}^{\infty}U(k)t^i\right)\right]_{t=0}, \quad k \geq 0. \tag{3.5.8}
$$

Example 10. Consider the initial value problem

$$u'' = 2u + 4u\ln u, \quad u(0) = 1,\ u'(0) = 0. \tag{3.5.9}$$

We first rewrite equation (3.5.9) as

$$u'' = 2u + f(u), \quad f(u) = u\ln u.$$

Applying the differential transform to the last equation, we have

$$(k+1)(k+2)U(k+2) - 2U(k) - 4F(k) = 0, \tag{3.5.10}$$

where $F(k)$ are the DT-components of the nonlinear term $f(u) = u\ln u$. From initial conditions we obtain

$$U(0) = 1, \quad U(1) = 0.$$

By applying formula for the nonlinear term $f(u)$, we obtain

$$\begin{aligned}
F(0) &= U(0)\ln U(0),\\
F(1) &= U(1)[1+\ln U(0),\\
F(2) &= U(2)[1+\ln U(0)] + \frac{U^2(1)}{2U(0)},\\
F(3) &= U(3)[1+\ln U(0)] + \frac{U(1)U(2)}{U(0)} - \frac{U^3(1))}{6U^2(0)},\\
&\vdots
\end{aligned}$$

In order to obtain the series solution, we use recurrence scheme (3.5.10) and the appropriate $F(k)$-components from equations (3.5.11). Hence

$$\begin{aligned}
&U(0) = 1, \quad U(1) = 0, \quad U(2) = 1,\\
&U(3) = 0, \quad U(4) = \tfrac{1}{2}, \quad U(5) = 0,\\
&U(6) = \tfrac{1}{3!}, \quad U(7) = 0, \quad U(8) = \tfrac{1}{4!},\\
&\vdots
\end{aligned}$$

and the series solution is readily obtained as

$$u = 1 + t^2 + \frac{t^4}{2!} + \frac{t^6}{3!} + \frac{t^8}{4!} + \frac{t^2 k}{k!} + \dots,$$

which is consistent for $k \to \infty$ with Maclaurin expansion of the exact solution $u = \mathrm{e}^{t^2}$.

Example 11. Consider the initial value problem of the relativistic harmonic oscillator

$$u'' + \frac{u}{\sqrt{1+u^2}} = 0, \quad u(0) = \alpha,\ u'(0) = 0. \tag{3.5.11}$$

As in previous example, equation (3.5.11) can be written in the form

$$u'' + f(u) = 0, \quad f(u) = \frac{u}{\sqrt{1+u^2}}. \tag{3.5.12}$$

Applying the differential transform to the last equation and the initial conditions, we have

$$(k+1)(k+2)U(k+2) - 2U(k) + F(k) = 0, \quad U(0) = \alpha,\ U(1) = 0. \tag{3.5.13}$$

By applying formula for the nonlinear term $f(u)$, we obtain

$$\begin{aligned}
F(0) &= \frac{U(0)}{\sqrt{1+U^2(0)}}, \\
F(1) &= \frac{U(1)}{[1+U^2(0)]^{3/2}}, \\
F(2) &= \frac{U(2)[1+U^2(0)] - 3U(0)U^2(1)}{2[1+U^2(0)]^{5/2}}, \\
F(3) &= \frac{1}{2[1+U^2(0)]^{7/2}} \\
&\times\ 2U(3)[1+U^2(0)]^2 - 6U(1)U(2)[U(0)+U^3(0)] + U^3(1)[4U^2(0)-1], \\
&\vdots
\end{aligned} \tag{3.5.14}$$

We obtain the following results for $k = 0, 1, 2, \ldots, 6$:

$$\begin{aligned}
U(0) &= \alpha, \quad U(1) = 0, \quad U(2) = \frac{-\alpha}{2\sqrt{1+\alpha^2}}, \\
U(3) &= 0, \quad U(4) = \frac{\alpha}{24(1+\alpha^2)^2}, \quad U(5) = 0, \\
U(6) &= \frac{-\alpha(1-9\alpha^2)}{720(1+\alpha^2)^{7/2}}, \quad U(7) = 0, \quad U(8) = \frac{\alpha - 99\alpha^3 + 180\alpha^5}{40320(1+\alpha^2)^5}.
\end{aligned}$$

Hence the series solution is given as

$$\begin{aligned}
u(t) &= \alpha - \left(\frac{\alpha}{2\sqrt{1+\alpha^2}}\right)t^2 + \left(\frac{\alpha}{24(1+\alpha^2)^2}\right)t^4 \\
&- \left(\frac{\alpha(1-9\alpha^2)}{720(1+\alpha^2)^{7/2}}\right)t^6 + \left(\frac{\alpha - 99\alpha^3 + 180\alpha^5}{40320(1+\alpha^2)^5}\right)t^8 + \ldots
\end{aligned}$$

The same series solution can be obtained using complex recurrence scheme but with very complicated recurrence equations. It is obvious that initial value problem (3.5.11) is unsolvable using classical DTM which is the main advantage of the presented general formula.

Example 12. Consider the special form of equation (3.4.3)

$$u'' + \frac{2}{t}u' + u^m = 0, \quad m \geq 0, \tag{3.5.15}$$

with initial conditions

$$u(0) = a, \quad u'(0) = 0. \tag{3.5.16}$$

Equation (3.5.15) is one of the basic equations in the theory of stellar structure. It is known that the polytropic theory of stars essentially follows out of thermodynamic considerations, that deal with the issue of energy transport, through the transfer of material between different levels of the star. Assume that the star is at hydrostatic equilibrium, then

$$\frac{dP}{dr} = -\rho\frac{gM(r)}{r^2}, \tag{3.5.17}$$

and the mass profile is given by

$$\frac{dM(r)}{dr} = 4\pi\rho r^2, \tag{3.5.18}$$

where g is the gravitational constant, P is the pressure at radius r, $M(r)$ is the mass of a star at a radius r, and ρ is the density at a distance r from the center of a spherical star. The polytropic assumption is given by

$$P = K\rho^{1+\frac{1}{m}}, \tag{3.5.19}$$

where K is a constant and m is the polytropic index which is related to the ratio of specific heats of the gas comprising the star. Differentiating equation (3.5.17) with respect r and substituting into equation (3.5.18) we get

$$\frac{1}{r^2}\frac{d}{dr}\left(\frac{r^2}{\rho}\frac{dP}{dr}\right) = -4\pi g\rho. \tag{3.5.20}$$

The additional alteration to the expression for density has been inserted with λ representing the central density of the star and u that of a related dimensionless quantity that are both related to through the following relation

$$\rho = \lambda u^m, \tag{3.5.21}$$

where m is a constant. Substituting (3.5.19), (3.5.21) into (3.5.20) we have

$$\left[\frac{K(m+1)}{4\pi g}\lambda^{\frac{1}{m}-1}\right]\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{du}{dr}\right) = -u^m. \tag{3.5.22}$$

Equation (3.5.22) can be simplified by the introduction of an additional dimensionless variable x, given by the following

$$r = ax, \quad a = \left[\frac{K(m+1)}{4\pi g}\lambda^{\frac{1}{m}-1}\right]^{\frac{1}{2}}.$$

Substituting these relations into equation (3.5.22) we obtain the known Lane-Emden type equation

$$\frac{1}{x^2}\frac{d}{dx}\left(x^2\frac{du}{dx}\right) = -u^m$$

which can be easily rewritten into the form (3.5.15). The physically interesting range of m is $0 \le m \le 5$. It has been claimed in the literature that only for $m = 0, 1$ and 5 the solutions of the Lane-Emden equation (also called the polytropic differential equations) are given in closed form. This equation describes the temperature variation of a spherical gas cloud under the mutual attraction of its molecules and subject to the laws of classical thermodynamics. It also describes the variation of density as a function of the radial distance for a polytrope.

Efficiency and simplicity of IDTM in comparison with current approach of DTM will be demonstrated with the following singular initial problem:

Consider polytropic equation (3.5.15) with $m = 5$, i.e.

$$u'' + \frac{2}{t}u' + u^5 = 0 \tag{3.5.23}$$

subject to initial conditions (3.5.16). At first, we will solve equation (3.5.23) by means of current approach of using DTM (see Theorem 3.1).

Multiplying both sides of equation (3.5.23) by t and then taking differential transformation we obtain the following relation

$$\sum_{l=0}^{k}\delta(l-1)(k+2-l)(k+1-l)U(k+2-l)+2(k+1)U(k+1)$$
$$+\sum_{l=0}^{k}\sum_{s=0}^{k-l}\sum_{m=0}^{k-l-s}\sum_{p=0}^{k-l-s-m}\sum_{q=0}^{k-l-s-m-p}\delta(l-1)$$
$$\times U(s)U(m)U(p)U(q)U(k-l-s-m-p-q)=0. \tag{3.5.24}$$

The recurrence relation has the form

$$U(k+1)=\frac{-1}{(k+1)(k+2)}\sum_{l=0}^{k}\sum_{s=0}^{k-l}\sum_{m=0}^{k-l-s}\sum_{p=0}^{k-l-s-m}\sum_{q=0}^{k-l-s-m-p}$$
$$\delta(l-1)U(s)U(m)U(p)U(q)U(k-l-s-m-p-q). \tag{3.5.25}$$

Using the initials conditions and recurrence relation (3.5.25) we get

$$U(0)=a,\ U(1)=0,\ U(2)=-\frac{a^5}{6},\ U(3)=0,\ U(4)=\frac{a^9}{24},\ U(5)=0,$$
$$U(6)=-\frac{5a^{13}}{432},\ U(7)=0,\ U(8)=\frac{35a^{17}}{10368},\ U(9)=0,\ldots$$

Hence, by the inverse transformation rule, we have

$$u(t)=a-\frac{a^5t^2}{6}+\frac{a^9t^4}{24}-\frac{5a^{13}}{432}t^6+\frac{35a^{17}t^8}{10368}-\ldots$$

and the closed form of the solution has the form

$$u(t)=\frac{a}{\sqrt{1+\frac{(a^2x)^2}{3}}}.$$

Now, we will solve equation (3.5.23) using IDTM based on differential transformation components (3.5.8).

Multiplying both sides of equation (3.5.23) by t and then taking differential transformation, we get

$$\sum_{l=0}^{k}\delta(l-1)(k+2-l)(k+1-l)U(k+2-l)+2(k+1)U(k+1)+\sum_{l=0}^{k}\delta(l-1)F(k-l)=0.$$

Thus

$$(k+1)kU(k+1)+2(k+1)U(k+1)+F(k-1)=0. \tag{3.5.26}$$

From (3.5.26), we get recurrence equations

$$U(k+1)=-\frac{F(k-1)}{(k+1)(k+2)},k\geq 1. \tag{3.5.27}$$

We have very simple recurrence equations in comparison with recurrence equations (3.5.25) obtained using basic formulas in Theorem 3.1.

From initial conditions, we have $U(0) = a$, $U(1) = 0$ and from (3.5.8), we have $F(0) = f(U(0)) = U(0)^5 = a^5$.
Hence, for $k = 1$ in (3.5.27), we obtain

$$\begin{aligned}
U(2) &= -\frac{F(0)}{6} = -\frac{f(U(0))}{6} = -\frac{U(0)^5}{6} = -\frac{a^5}{6}, \\
U(3) &= -\frac{F(1)}{12} = -\frac{U(1)f'(U(0))}{12} = 0, \\
U(4) &= -\frac{F(2)}{20} = -\frac{1}{20}\left[U(2)f'(U(0)) + \frac{1}{2}U^2(1)f''(U(0))\right] = -\frac{1}{20}\left(-\frac{a^5}{6}\right)5U(0)^4 = \frac{a^9}{24}, \\
&\vdots
\end{aligned}$$

We obtained the same result as in the case of the application of current approach of using DTM but with very simple recurrence equations.

Remark 1. From (3.5.8), it is obvious that we can solve polytropic equation (3.5.15) for any $m \in \mathbb{R}$ whereas current approach of using DTM (see the last formula in Theorem 3.1) can be applied only for $m \in \mathbb{N} \cup \{0\}$.

Example 13. Consider an isothermal gas sphere where the pressure P and the spatial density ρ are related as follows:

$$P = K\rho + D \tag{3.5.28}$$

where K and D are constants which depend on the thermodynamic properties of the isothermal gas sphere. Let r denote the radius measured from the center of the configuration. Since we consider in particular isothermal gas spheres we are concerned with spherical symmetrical distributions of matter. Hence the total pressure P the spatial density ρ and other physical variables will be functions of r only. Consider the substitutions

$$\rho = \rho_c \mathrm{e}^{-u}, \quad r = \left[\frac{K}{4\pi G\rho_c}\right]^{1/2} x, \tag{3.5.29}$$

where ρ_c is defined to be central density. It is noted that the central density is also the maximum density at $r = 0$ since the density decayes from the center of the star in an outwardly direction as indicated by the substitution $\rho = \rho_c \mathrm{e}^{-u}$. Substituting (3.5.28) into (3.5.17) and (3.5.18) and using the transformations (3.5.29), differential equation (3.5.20) has the form

$$\frac{1}{x^2}\frac{d}{dx}\left(x^2\frac{du}{dx}\right) = \mathrm{e}^{-u} \tag{3.5.30}$$

which describes the dimensionless density distribution in an isothermal gas sphere. Equation (3.5.30) can be written as follows:

$$u'' + \frac{2}{x}u' - \mathrm{e}^{-u} = 0. \tag{3.5.31}$$

It is known that

$$u = -\ln\left(\frac{2}{x^2}\right) \tag{3.5.32}$$

is one of the solutions of equation (3.5.31). This solution describes a model known as the singular isothermal sphere which has infinite density at $x = 0$. To obtain a solution that is well behaved at the origin, equation (3.5.31) has to be integrated with the central conditions

$$u(0) = 0, \quad u'(0) = 0. \tag{3.5.33}$$

In the following will be shown that singular initial value problem (3.5.31),(3.5.33) has only approximate solution.

Consider singular initial value problem (3.5.31), (3.5.33). In our case the nonlinear term has the form $f(u) = \mathrm{e}^{-u}$. Applying the differential transformation to the both sides of the last equation, we get

$$(k+1)F(k+1) = -\sum_{l=0}^{k}(l+1)F(k-l)U(l+1). \tag{3.5.34}$$

At first, we have

$$F(0) = \mathrm{e}^{-U(0)}.$$

For $k = 0$ in (3.5.34), we obtain

$$F(1) = -U(1)F(0) = -U(1)\mathrm{e}^{-U(0)}.$$

For $k = 1$ in (3.5.34)), we have

$$F(2) = -\frac{1}{2}\left[F(1)U(1) + 2F(0)U(2)\right] = -\left[U(2) + \frac{U^2(1)}{2}\right]\mathrm{e}^{-U(0)}.$$

Similarly, for $k = 2$, we get

$$F(3) = \left[U(3) + U(1)U(2) + \frac{U^3(1)}{3!}\right]\mathrm{e}^{-U(0)}.$$

In the same way we can obtain the differential transformation components $F(4), F(5), \ldots$

Multiplying both sides of equation (3.5.31) by t and then taking differential transformation, we get

$$\sum_{l=0}^{k}\delta(l-1)(k+2-l)(k+1-l)U(k+2-l) + 2(k+1)U(k+1) - \sum_{l=0}^{k}\delta(l-1)F(k-l) = 0.$$

Hence

$$U(k+1) = \frac{F(k-1)}{(k+1)(k+2)}, k \geq 1. \tag{3.5.35}$$

We know, from initial conditions (3.5.33), that $U(0) = U(1) = 0$. Then

$$\begin{aligned}
U(2) &= \frac{F(0)}{6} = \frac{\mathrm{e}^{-U(0)}}{6} = \frac{1}{6}, \\
U(3) &= \frac{F(1)}{12} = -\frac{1}{12}U(1)\mathrm{e}^{-U(0)} = 0, \\
U(4) &= \frac{F(2)}{20} = -\frac{1}{20}\left[U(2) + \frac{U_1^2(1)}{2}\right]\mathrm{e}^{-U(0)} = -\frac{1}{120},
\end{aligned}$$

$$U(5) = \frac{F(3)}{30} = -\frac{1}{30}\left[U(3) + U(1)U(2) + \frac{U^3(1)}{3!}\right] \mathrm{e}^{-U(0)} = 0,$$
$$\vdots$$

Then the series solution of (3.5.31), (3.5.33) can be written as follows:

$$u(t) = \frac{1}{6}t^2 - \frac{1}{5\cdot 4!}t^4 + \frac{8}{21\cdot 6!}t^6 - \frac{122}{81\cdot 8!}t^8 + \frac{4087}{495\cdot 10!} - \dots$$

The same initial problem has been solved by Heydari et al. [6], Wazwaz [5], Singh et al. [7] using radial basis functions, Adomian decomposition method and homotopy perturbation method respectively. The following table shows (for $N = 10$) the comparison of $u(t)$ obtained by the presented method (IDTM) and those obtained by Heydari, Wazwaz and Singh.

t	IDTM	Heydari [6]	Wazwaz [5]	Singh [7]
0.0	0.0000000000	0.0000000000	0.0000000000	0.0000000000
0.1	0.0016658338	0.0016658328	0.0016658338	0.0016658338
0.2	0.0066533671	0.0066533691	0.0066533671	0.0066533671
0.5	0.0411539568	0.0411539552	0.0411539568	0.0411539568
1.0	0.1588273536	0.1588276770	0.1588273536	0.1588273536

From the table it is obvious that differential transformation method using very simple recurrence scheme gives the same results as Adomian decomposition method and homotopy perturbation where it is necessary to calculate symbolic derivatives and multiple integrals and make complicated calculations of the Adomian polynomials. Moreover, using both mentioned methods an initial guess estimation is required.

3.6 DTM for Functional Differential Equations with Proportional Delay

In this section, we consider the following nonlinear differential and integro-differential equations with proportional delays:

$$\mathcal{F}\Big(t, \mathbf{U}(pt)\Big) = 0, \quad t \geq 0, \tag{3.6.1}$$

$$\mathcal{G}\left(t, \mathbf{U}(pt), \int_0^{rt} K\big(t, s, \mathbf{U}(qt)\big)ds\right) = 0, \quad t \geq 0, \tag{3.6.2}$$

where $\mathcal{F}, \mathcal{G}, K$ are given functions with appropriate domains of definition,

$$\mathbf{U}(pt) = (u(p_0t), u'(p_1t), \dots, u^{(n)}(p_nt)), \mathbf{U}(qt) = (u(q_0s), u'(q_1s), \dots, u^{(m)}(q_ms)),$$

$p_i,\ q_j,\ r \in (0,1),\ i = 0, 1, \dots, n,\ j = 0, 1, \dots, m,\ m < n.$

Functional differential and integro-differential equations with proportional delays are usually referred to as pantograph equations or generalized equations and are often used to model of some problems with aftereffect in mechanics and the related scientific fields as well. Many typical applications of functional differential equations such as models for stress-strain states of materials, motion of rigid bodies and models of polymer crystallization can be found in Kolmanovskii and Myshkis's monograph [9] and the references therein.

At first, we state the fundamental transformation formulas for solving equations (3.6.1), (3.6.2):

Theorem 3.4 ([8]). *Assume that $W(k), U(k)$ and $U_i(k)$ are the differential transformations of the functions $w(t), u(t)$ and $u_i(t)$, respectively, and $q, q_i, \in (0,1)$, $i = 1, 2$. Then:*

(i) If $w(t) = u(qt)$, then $W(k) = q^k U(k)$.

(ii) If $w(t) = u_1(q_1 t) u_2(q_2 t)$, then $W(k) = \sum_{l=0}^{k} q_1^l q_2^{k-l} U_1(l) U_2(k-l)$.

(iii) If $w(t) = \frac{d^m}{dt^m} u(qt)$, then $W(k) = \frac{(k+m)!}{k!} q^{k+m} U(k+m)$.

(iv) If $w(t) = \frac{d^n}{dt^n} u_1(q_1 t) \frac{d^m}{dt^m} u_2(q_2 t)$, then

$$W(k) = \sum_{l=0}^{k} q_1^{l+n} q_2^{k-l+m} \frac{(l+n)!(k-l+m)!}{l!(k-l)!} U_1(l+n) U_2(k-l+m).$$

Proof. (i) From equation (3.1.2), we get

$$\frac{d^k}{dt^k} w(t) = \frac{d^k}{dt^k} [uqt)] = q^k \frac{d^k}{d\tilde{t}^k} u(\tilde{t}),$$

where $\tilde{t} = qt$, thus

$$\left[\frac{d^k}{dt^k} w(t)\right]_{t=t_0} = q^k \left[\frac{d^k}{d\tilde{t}^k} u(\tilde{t})\right]_{t=t_0} = q^k k! U(k)$$

and using (3.1.1) we have

$$W(k) = \frac{1}{k!} \left[\frac{d^k w(t)}{dt^k}\right]_{t=t_0} = \frac{1}{k!} q^k k! U(k) = q^k U(k),$$

where $k \in \mathbb{N} \cup \{0\}$.

(ii) Using Leibnitz formula, we obtain

$$\begin{aligned} \frac{d^k}{dt^k} w(t) &= \frac{d^k}{dt^k} \left[u_1(q_1 t) u_2(q_2 t)\right] = \sum_{l=0}^{k} \binom{k}{l} \frac{d^l}{dt^l} \left[u_1(q_1 t)\right] \frac{d^{k-l}}{dt^{k-l}} \left[u_2(q_2 t)\right] \\ &= \sum_{l=0}^{k} \binom{k}{l} \frac{d^l}{d\tilde{t}^l} \left[u_1(\tilde{t})\right] \frac{d^{k-l}}{d\hat{t}^{k-l}} \left[u_2(\hat{t})\right], \end{aligned}$$

where $\tilde{t} = q_1 t$, $\hat{t} = q_2 t$, hence

$$\left[\frac{d^k}{dt^k} w(t)\right]_{t=t_0} = \sum_{l=0}^{k} \binom{k}{l} \left[q_1^l l! U_1(l)\right] \left[q_2^{k-l} (k-l)! U_2(k-l)\right] = \sum_{l=0}^{k} k! q_1^l q_2^{k-l} U_1(l) U_2(k-l).$$

From equation (3.1.1), we have

$$W(k) = \frac{1}{k!} \left[\frac{d^k}{dt^k} w(t)\right]_{t=t_0} = \sum_{l=0}^{k} k! q_1^l q_2^{k-l} U_1(l) U_2(k-l), \quad k \in \mathbb{N} \cup \{0\}.$$

(iii) From the proof of formula (i), we get

$$\left[\frac{d^{k+m}}{dt^{k+m}} w(t)\right]_{t=t_0} = q^{k+m} \left[\frac{d^{k+m}}{d\tilde{t}^{k+m}} u(\tilde{t})\right]_{t=t_0} = (k+m)! q^{k+m} U(k+m).$$

Then

$$W(k) = \frac{1}{k!}\left[\frac{d^k}{dt^k}w(t)\right]_{t=t_0} = \frac{(k+m)!}{k!}q^{k+m}U(k+m).$$

(iv) Analogously from to previous part of the proof, we get

$$\begin{aligned}
\frac{d^k}{dt^k}w(t) &= \frac{d^k}{dt^k}\left[\frac{d^n}{dt^n}u_1(q_1t)\frac{d^m}{dt^m}u_2(q_2t)\right] \\
&= \sum_{l=0}^{k}\binom{k}{l}\frac{d^l}{dt^l}\left[\frac{d^n}{dt^n}u_1(q_1t)\right]\frac{d^{k-l}}{dt^{k-l}}\left[\frac{d^m}{dt^m}u_2(q_2t)\right] \\
&= \sum_{l=0}^{k}\binom{k}{l}q_1^{l+n}\frac{d^{l+n}}{d\tilde{t}^{l+n}}u_1(\tilde{t})q_2^{k-l+m}\frac{d^{k-l+m}}{d\hat{t}^{k-l+m}}u_2(\hat{t}),
\end{aligned}$$

where $\tilde{t} = q_1t,\ \hat{t} = q_2t$, therefore

$$\begin{aligned}
\frac{d^k}{dt^k}w(t) &= \sum_{l=0}^{k}\binom{k}{l}\left[q_1^{l+n}(l+n)!U_1(l+n)\right]\left[q_2^{k-l+m}(k-l+m)!U_2(k-l+m)\right] \\
&= \sum_{l=0}^{k}\frac{k!(l+n)!(k-l+m)!}{l!(k-l)!}q_1^{l+n}q_2^{k-l+m}U_1(l+n)U_2(k-l+m).
\end{aligned}$$

Then from (3.1.1), we have

$$W(k) = \sum_{l=0}^{k}q_1^{l+n}q_2^{k-l+m}\frac{(l+n)!(k-l+m)!}{l!(k-l)!}U_1(l+n)U_2(k-l+m).$$

The proof is complete.

□

Theorem 3.5 ([8]). *Assume that $W(k), U(k)$ and $U_i(k)$ are the differential transformations of the functions $w(t), u(t)$ and $u_i(t)$, respectively, and $r, q, q_i \in (0,1),\ i = 1,2$. Then*

$$\begin{aligned}
(I)\quad \text{If } w(t) &= \int_0^{rt}u(qs)ds, \text{ then } W(k) = \frac{1}{k}r^kq^{k-1}U(k-1). \\
(II)\quad \text{If } w(t) &= \int_0^{rt}u_1(q_1s)u_2(q_2s)ds, \text{ then } W(k) = \frac{1}{k}\sum_{l=0}^{k-1}r^kq_1^lq_2^{k-l-1}U_1(l)U_2(k-l-1). \\
(III)\ \text{If } w(t) &= u(qt)\int_0^{rt}u_1(q_1s)u_2(q_2s)ds, \text{ then} \\
W(k) &= \sum_{l=0}^{k-1}\sum_{s=0}^{k-l-1}\frac{1}{k-l}r^{k-l}q^lq_1^sq_2^{k-l-s-1}U(l)U_1(s)U_2(k-l-s-1),
\end{aligned}$$

where $k \in \mathbb{N}$.

Proof. (I) It is obvious that

$$\frac{d^k}{dt^k}w(t) = r\frac{d^{k-1}}{dt^{k-1}}u(rqt),$$

then

$$\left[\frac{d^k}{dt^k}w(t)\right]_{t=t_0} = r(k-1)!(rq)^{k-1}U(k-1) = (k-1)!r^kq^{k-1}U(k-1).$$

From here and equation (3.1.1), we get

$$W(k) = \frac{1}{k!}\left[\frac{d^k}{dt^k}w(t)\right]_{t=t_0} = \frac{1}{k}r^kq^{k-1}U(k-1), \quad k \in \mathbb{N}.$$

(II) Similarly as in previous part (I), we have

$$\frac{d^k}{dt^k}w(t) = r\frac{d^{k-1}}{dt^{k-1}}\left[u_1(rq_1t)u_2(rq_2t)\right] = r\sum_{l=0}^{k-1}\binom{k-1}{l}(rq_1)^l\frac{d^l}{d\tilde{t}^l}u_1(\tilde{t})(rq_2)^{k-l-1}\frac{d^{k-l-1}}{d\hat{t}^{k-l-1}}u_2(\hat{t}),$$

where $\tilde{t} = q_1t,\ \hat{t} = q_2t$. Then

$$\left[\frac{d^k}{dt^k}w(t)\right]_{t=t_0} = r\sum_{l=0}^{k-1}\binom{k-1}{l}(rq_1)^ll!U_1(l)(rq_2)^{k-l-1}(k-l-1)!U_2(k-l-1).$$

Using equation (3.1.1), we obtain

$$W(k) = \frac{1}{k!}\left[\frac{d^k}{dt^k}w(t)\right]_{t=t_0} = \frac{1}{k}\sum_{l=0}^{k-1}r^kq_1^lq_2^{k-l-1}U_1(l)U_2(k-l-1), \quad k \in \mathbb{N}.$$

(III) Put $h(t) = \int_0^{rt} u_1(q_1s)u_2(q_2s)ds$, then from previous part, we get

$$\frac{d^k}{dt^k}w(t) = \frac{d^k}{dt^k}\left[u(qt)h(t)\right] = \sum_{l=0}^{k}\binom{k}{l}p^l\frac{d^l}{d\tilde{t}^l}u(\tilde{t})\frac{d^{k-l}}{dt^{k-l}}h(t), \tag{3.6.3}$$

where $\tilde{t} = qt$, and

$$\begin{aligned}\frac{d^{k-l}}{dt^{k-l}}h(t) &= r\frac{d^{k-l-1}}{dt^{k-l-1}}\left[u_1(rq_1t)u_2(rq_2t)\right]\\ &= r\sum_{s=0}^{k-l-1}\binom{k-l-1}{s}(rq_1)^s\frac{d^s}{d\tilde{t}^s}u_1(\tilde{t})(rq_2)^{k-l-s-1}\frac{d^{k-l-s-1}}{d\hat{t}^{k-l-s-1}}u_2(\hat{t}),\end{aligned} \tag{3.6.4}$$

where $\tilde{t} = q_1t,\ \hat{t} = q_2t$. From (3.6.3) and (3.6.4), we obtain

$$\begin{aligned}\left[\frac{d^k}{dt^k}w(t)\right]_{t=t_0} &= \sum_{l=0}^{k}\sum_{s=0}^{k-l-1}\binom{k}{l}\binom{k-l-1}{s}r^{k-l}q^lq_1^sq_2^{k-l-s-1}\\ &\times\ l!s!(k-l-s-1)!U(l)U_1(s)U_2(k-l-s-1),\end{aligned}$$

but for $l = k$ we get

$$\left[\frac{d^{k-l}}{dt^{k-l}}y(t)\right]_{t=t_0} = 0.$$

Then, using equation (3.1.1), we get

$$W(k) = \sum_{l=0}^{k-1} \sum_{s=0}^{k-l-1} \frac{1}{k-l} r^{k-l} q^l q_1^s q_2^{k-l-s-1} U(l) U_1(s) U_2(k-l-s-1), \quad k \in \mathbb{N}.$$

The proof is complete.

□

Example 14. To show a practical application of derived formulas, we consider the following pantograph delay equation:

$$u'(t) = \frac{1}{2} \exp\left(\frac{t}{2}\right) u\left(\frac{t}{2}\right) + \frac{1}{2} u(t), \quad u(0) = 1. \tag{3.6.5}$$

Using differential transformation method, the differential transform version of equation (3.6.5) is

$$(k+1)U(k+1) = \frac{1}{2} \sum_{l=0}^{k} \frac{1}{2^l l!} \frac{1}{2^{k-l}} U(k-l) + \frac{1}{2} U(k), \quad k \geq 0, \tag{3.6.6}$$

and the differential transform version of the initial condition $u(0) = 1$ has the form $U(0) = 1$. From equation (3.6.6), we obtain recurrence system of equations

$$U(k+1) = \frac{1}{2k+2} \left[\sum_{l=0}^{k} \frac{1}{2^k l!} U(k-l) + U(k) \right], \quad k \geq 0. \tag{3.6.7}$$

From system (3.6.7), we have

$$\begin{aligned}
U(1) &= \frac{1}{2}\big[U(0) + U(0)\big] = 1, \\
U(2) &= \frac{1}{4}\left[\frac{1}{2}U(1) + \frac{1}{2}U(0) + U(1)\right] = \frac{1}{2}, \\
U(3) &= \frac{1}{6}\left[\frac{1}{4}U(2) + \frac{1}{4}U(1) + \frac{1}{8}U(0) + U(2)\right] = \frac{1}{6}, \\
U(4) &= \frac{1}{8}\left[\frac{1}{8}U(3) + \frac{1}{8}U(2) + \frac{1}{16}U(1) + \frac{1}{48}U(0) + U(3)\right] = \frac{1}{24}, \\
&\vdots
\end{aligned}$$

Using the inverse transformation rule, we obtain the following series solution

$$u(t) = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \frac{t^4}{4!} + \cdots + \frac{t^k}{k!} + \ldots$$

The closed form of above series solution is $u(t) = \mathrm{e}^t$, which is the exact solution of equation (3.6.5).

Example 15. Consider the following delay differential equation of the third order:

$$u'''(t) = -1 + 2u^2\left(\frac{t}{2}\right), \quad u(0) = 0,\ u'(0) = 1,\ u''(0) = 0. \tag{3.6.8}$$

Using differential transformation method, the differential transform version of equation (3.6.8), we get

$$(k+3)(k+2)(k+1)U(k+3) = -\delta(k) + 2\sum_{l=0}^{k} \frac{1}{2^l}U(l)\frac{1}{2^{k-l}}U(k-l) \tag{3.6.9}$$

and the differential transformation of the initial conditions $u(0) = 0,\ u'(0) = 1,\ u''(0) = 0$ gives

$$U(0) = 0,\ U(1) = 1,\ U(2) = 0.$$

From (3.6.9) we have

$$U(k+3) = \frac{1}{(k+3)(k+2)(k+1)}\left[-\delta(k) + 2\sum_{l=0}^{k}\frac{1}{2^k}U(l)U(k-l)\right],\ k \geq 0. \tag{3.6.10}$$

Solving recurrence equations (3.6.10) we get

$$\begin{aligned}
U(3) &= \frac{1}{6}\big[-1 + 2U(0)^2\big] = -\frac{1}{6} = \frac{1}{3!},\\
U(4) &= \frac{1}{24}2\left[\frac{1}{2}U(0)U(1) + \frac{1}{2}U(1)U(0)\right] = 0,\\
U(5) &= \frac{1}{60}2\left[\frac{1}{4}U(0)U(2) + \frac{1}{4}U^2(1) + \frac{1}{4}U(0)U(2)\right] = \frac{1}{120} = \frac{1}{5!},\\
U(6) &= \frac{1}{120}2\left[\frac{1}{8}U(0)U(3) + \frac{1}{8}U(1)U(2) + \frac{1}{8}U(2)U(1) + \frac{1}{8}U(3)U(0)\right] = 0,\\
&\vdots
\end{aligned}$$

Thus

$$U(k) = \begin{cases} 0, & k = 2n,\\ \dfrac{(-1)^n}{(2n+1)!}, & k = 2n+1,\end{cases}$$

where $n \in \mathbb{N}\cup\{0\}$. From here and using the inverse transformation rule, we obtain series solution in the form

$$u(t) = \sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)!}t^n = \sin t,$$

which is the exact solution of equation (3.6.8).

Example 16. Consider the following delay differential equation of the second order:

$$u''(t) = u'(t) + 2u^2\left(\frac{t}{2}\right) - 3u\left(\frac{t}{3}\right)u\left(\frac{t}{2}\right) - 2t + 2, \quad u(0) = u'(0) = 0. \tag{3.6.11}$$

Applying differential transformation method to equation (3.6.11) we get

$$\begin{aligned}
(k+2)(k+1)U(k+2) &= (k+1)U(k+1) + 2\sum_{l=0}^{k}\frac{1}{2^l}U(l)\frac{1}{2^{k-l}}U(k-l)\\
&- 3\sum_{l=0}^{k}\frac{1}{3^l}U(l)\frac{1}{2^{k-l}}U(k-l) - 2\delta(k-1) + 2\delta(k),
\end{aligned} \tag{3.6.12}$$

and for initial conditions $u(0) = u'(0) = 0$, we have $U(0) = U(1) = 0$. From (3.6.12), we obtain recurrence equations

$$U(k+2) = \frac{1}{k+2}U(k+1) + \frac{1}{(k+2)(k+1)} \times \left[2\sum_{l=0}^{k}\frac{1}{2^k}U(l)U(k-l) - 3\sum_{l=0}^{k}\frac{1}{3^l}\frac{1}{2^{k-l}}U(l)U(k-l) - 2\delta(k-1) + 2\delta(k)\right]. \quad (3.6.13)$$

Solving recurrence equations (3.6.13), we have

$$\begin{aligned}
U(2) &= \frac{1}{2}U(1) + \frac{1}{2}\left[2U^2(0) - 3U^2(0) + 2\right] = 1,\\
U(3) &= \frac{1}{3}U(2) + \frac{1}{6}\left[2\big(\frac{1}{2}U(0)U(1) + \frac{1}{2}U(1)U(0)\big) - 3\big(\frac{1}{2}U(0)U(1) + \frac{1}{3}U(1)U(0)\big) - 2\right] = 0,\\
U(4) &= \frac{1}{4}U(3) + \frac{1}{12}\left[U(0)U(2) + \frac{1}{2}U^2(1) - 3\big(\frac{1}{4}U(0)U(2) + \frac{1}{6}U^2(1) + \frac{1}{9}U(2)U(0)\big)\right] = 0,\\
&\vdots
\end{aligned}$$

Thus $U(k) = 0,\ k \geq 3$. Using the inverse transformation rule, we obtain the exact solution $u(t) = t^2$.

Example 17. Consider nonlinear pantograph-type integro-differential equation of the first order:

$$u'(t) + \left(\frac{1}{2}t - 2\right)u(t) - 2\int_0^t u^2\left(\frac{s}{2}\right)ds = 1, \quad (3.6.14)$$

subject to initial condition $u(0) = 0$. Substituting $t = 0$ in integro-differential equation (3.6.14), we get $u'(0) = 1$. Then the differential transformed version of equation (3.6.14) has the form

$$(k+1)U(k+1) + \sum_{l=0}^{k}\big(\frac{1}{2}\delta(l-1) - 2\delta(l)\big)U(k-l) - \frac{2}{k}\sum_{l=0}^{k-1}\frac{1}{2^{k-1}}U(l)U(k-l-1) = \delta(k). \quad (3.6.15)$$

Solving recurrence equations (3.6.15), we get

$$\begin{aligned}
&2U(2) - 2U(1) + \frac{1}{2}U(0) - 2U^2(0) = 0 \Rightarrow U(2) = 1,\\
&3U(3) + \left[-2U(2) + \frac{1}{2}U(1)\right] - U(0)U(1) = 0 \Rightarrow U(3) = \frac{1}{2!},\\
&4U(4) + \left[-2U(3) + \frac{1}{2}U(2)\right] - \frac{1}{6}\left[2U(0)U(2) + U^2(1)\right] = 0 \Rightarrow U(4) = \frac{1}{6} = \frac{1}{3!},\\
&5U(5) + \left[-2U(4) + \frac{1}{2}U(3)\right] - \frac{1}{8}\left[U(0)U(3) + U(1)U(2)\right] = 0 \Rightarrow U(5) = \frac{1}{24} = \frac{1}{4!},\\
&\vdots
\end{aligned}$$

From here and using the inverse transformation rule (3.1.2), we obtain series solution in the form

$$u(t) = t + t^2 + \frac{1}{2!}t^3 + \frac{1}{3!}t^4 + \frac{1}{4!}t^5 + \cdots + \frac{1}{k!}t^{k+1} + \cdots$$

The closed form of above series solution is $u(t) = te^t$, which is the exact solution of equation (3.6.14).

Example 18. Consider the following nonhomogeneous first-order integro-differential equation with proportional delay:

$$u'(t) - u\left(\frac{t}{2}\right) - \frac{1}{2}u\left(\frac{t}{2}\right)\int_0^{\frac{t}{3}} u(s)u\left(\frac{s}{2}\right)ds = 4 - 2t - \frac{8}{81}t^4, \tag{3.6.16}$$

subject to initial condition $u(0) = 0$. Substituting $t = 0$ in equation (3.6.16) we obtain the second condition $u'(0) = 4$. Now, applying differential transformation method to equation (3.6.16), we get

$$\begin{aligned}(k+1)U(k+1) &- \frac{1}{2^k}U(k) - \frac{1}{2}\sum_{l=0}^{k-1}\sum_{s=0}^{k-l-1}\frac{1}{k-l}\frac{1}{3^{k-l}}\frac{1}{2^{k-s-1}}U(l)U(s)U(k-l-s-1)\\ &= 4\delta(k) - 2\delta(k-1) - \frac{8}{81}t^4,\end{aligned}$$

and for initial conditions $u(0) = 0,\ u'(0) = 4$, we have $U(0) = 0,\ U(1) = 4$. Using the same procedure as in the above mentioned examples we get

$$\begin{aligned}&2U(2) - \frac{1}{2}U(1) - \frac{1}{6}U^3(0) = -2 \Rightarrow U(2) = 0,\\ &3U(3) - \frac{1}{4}U(2) - \frac{1}{2}\big[\frac{1}{18}U^2(0)U(1) + \frac{1}{9}U(0)U(1)U(0) + \frac{1}{6}U^2(1)U(0)\big] = 0 \Rightarrow U(3) = 0,\\ &\vdots\end{aligned}$$

Similarly we obtain $U(k) = 0,\ k \geq 2$. Using the inverse transformation rule, we get the exact solution $u(t) = 4t$.

3.7 DTM for Functional Differential Equations with Constant Delay

In this section we focus on functional differential equations with constant delays, i. e., equations of delayed or neutral type. The simplest delayed differential equation (DDE) with constant delay can have a form

$$x'(t) = x(t-1), \tag{3.7.1}$$

while simple neutral differential equation (NDE) can look like

$$x'(t) = x'(t-1). \tag{3.7.2}$$

To complete the list of possible simple forms, we mention equation

$$x'(t-1) = x(t), \tag{3.7.3}$$

which can be transformed ($t = \hat{t} + 1$) to equation

$$x'(\hat{t}) = x(\hat{t} + 1). \tag{3.7.4}$$

Equation (3.7.4) is a differential equation of advanced type (advanced differential equation, ADE) which will not be considered in this text.

In the most general form, we will consider the following functional differential equation (DDE or NDE) of n-th order with multiple constant delays

$$u^{(n)}(t) = f(t, u(t), u'(t), \dots, u^{(n-1)}(t), \mathbf{u}_1(t-\tau_1), \mathbf{u}_2(t-\tau_2), \dots, \mathbf{u}_r(t-\tau_r)), \tag{3.7.5}$$

where $\mathbf{u}_i(t-\tau_i) = (u(t-\tau_i), u'(t-\tau_i), \dots, u^{(m_i)}(t-\tau_i))$ is m_i-dimensional vector function, $m_i \le n$, $i = 1, 2, \dots, r$, $r \in \mathbb{N}$ and $f\colon [t_0, \infty) \times R^n \times R^\omega$ is a continuous function, where $\omega = \sum_{i=1}^{r} m_i$. Denote $t^* = \max\{\tau_1, \tau_2, \dots, \tau_r\}$, $m = \max\{m_1, m_2, \dots, m_r\}$. Obviously $m \le n$. In case $m = n$ equation (3.7.5) is of neutral type, otherwise it is DDE.

Why do we consider equations in the form (3.7.5)? Investigation of such equations is important since there is plenty of applications of these equations in real life. To name a few examples, we mention models describing behaviour of the central nervous system in a learning process, species populations struggling for a common food, dynamics of an autogenerator with delay and second-order filter, systems controlled by PI or PID regulators, evolution of population of one species etc. For further models and details, see e.g. [9] or [10].

For an instructive and clear example, we refer to one particular application, which is an example of showering person described in monograph [9], pp. 19–22.

What do we need to define the problem properly? We need equation (3.7.5) and we need some "initial conditions". In case of DDE and NDE, it appears that initial *function*, denote it $\phi(t)$, is needed to be assigned for equation (3.7.5) on the interval $[t_0 - t^*, t_0]$. For the sake of simplicity, we assume that $\phi(t) \in C^n([t_0 - t^*, t_0])$.

The crucial idea of the algorithm is to combine differential transformation method and general method of steps (more details on method of steps can be found for instance in monographies Kolmanovskii and Myshkis [9], Hale and Verduyn Lunel [10] or Bellen and Zennaro [14]). This approach enables us to replace the terms involving delay with initial function and its derivatives. Consequently, the original Cauchy problem for delayed or neutral differential equation is reduced to Cauchy problem for ordinary differential equation and further, using DTM, to a system of recurrence algebraic relations.

3.7.1 Algorithm description

Consider equation (3.7.5) subject to initials conditions

$$u(t_0) = u_0, u'(t_0) = u_1, \dots, u^{(n-1)}(t_0) = u_{n-1} \tag{3.7.6}$$

and subject to initial function $\phi(t)$ on interval $[t_0 - t^*, t_0]$ such that

$$\phi(t_0) = u(t_0), \phi'(t_0) = u'(t_0), \dots, \phi^{(n-1)}(t_0) = u^{(n-1)}(t_0). \tag{3.7.7}$$

First we apply the method of steps. We substitute the initial function $\phi(t)$ and its derivatives in all places where unknown functions with deviating arguments and derivatives of that functions appear. Then equation (3.7.5) changes to ordinary differential equation

$$u^{(n)}(t) = f(t, u(t), u'(t), \dots, u^{(n-1)}(t), \mathbf{\Phi}_1(t-\tau_1), \mathbf{\Phi}_2(t-\tau_2), \dots, \mathbf{\Phi}_r(t-\tau_r)),$$

where $\mathbf{\Phi}_i(t-\tau_i) = (\phi(t-\tau_i), \phi'(t-\tau_i), \dots, \phi^{(m_i)}(t-\tau_i))$, $m_i \le n$, $i = 1, 2, \dots, r$. Now applying DTM we get recurrence equation

$$\frac{(k+n)!}{k!} U(k+n) = \mathcal{F}\Big(k, U(0), U(1), \dots, U(k+n-1)\Big),$$

where, in general, $\mathcal{F}$ is a nonlinear function of its arguments.
Using transformed initial conditions and then inverse transformation rule, we obtain approximate solution of equation (3.7.5) in the form of infinite Taylor series

$$u(t) = \sum_{k=0}^{\infty} U(k)(t-t_0)^k$$

on the interval $[t_0, t_0+\alpha]$, where $\alpha = \min\{\tau_1, \tau_2, \ldots, \tau_r\}$, and $u(t) = \phi(t)$ on the interval $[t_0-t^*, t_0]$. It is obvious that since the conditions (3.7.7) are valid, the solution $u(t)$ must generally be in the class $C^{n-1}([t_0-t^*, t_0+\alpha])$.

3.7.2 Applications

Example 19. Consider equation

$$u''(t) - tu'(t-1) + u(t-2) = -t^2 - 2t + 5. \tag{3.7.8}$$

Recall that to formulate Cauchy problem for (3.7.8) correctly, we have to prescribe initial function $\phi(t)$ on $[-2, 0]$ satisfying conditions (3.7.7) as well:

$$u(t) = \phi(t) \text{ for } t \in [-2, 0),\ u(0) = \phi(0) = u_0,\ u'(0) = \phi'(0) = u_1. \tag{3.7.9}$$

Thus a solution of Cauchy problem (3.7.8), (3.7.9) should be expected in the form

$$u(t) = \begin{cases} \phi(t), & t \in [-2, 0], \\ \psi(t), & t \in [0, 1]. \end{cases}$$

For initial function $\phi(t) = t^2 - 1$ we have

$$u(t) = \phi(t) = t^2 - 1 \text{ for } t \in [-2, 0),\ u(0) = \phi(0) = -1,\ u'(0) = \phi'(0) = 0. \tag{3.7.10}$$

We use the above described algorithm to find unique solution of Cauchy problem (3.7.8), (3.7.10) on $[0, 1]$.

In first step, we substitute $\phi(t-2)$ for $u(t-2)$ and $\phi'(t-1)$ for $u'(t-1)$:

$$u''(t) - t \cdot 2(t-1) + \left[(t-2)^2 - 1\right] = -t^2 - 2t + 5.$$

After some rearrangements we obtain equation

$$u''(t) = 2.$$

Then in second step we apply DTM and we get simple recurrence relation

$$U(k+2) = \frac{2\delta(k)}{(k+1)(k+2)},\ k \geq 0.$$

From this relation and the fact that $U(0) = -1$, $U(1) = 0$, we get $U(2) = 1$, $U(k) = 0$ for $k \geq 3$. Hence the unique exact solution of Cauchy problem (3.7.8), (3.7.10) on $[0, 1]$ is $u(t) = t^2 - 1$. However, it can be easily verifed that, in fact, it is a solution on $[0, \infty)$.

Example 20. Consider again equation

$$u''(t) - tu'(t-1) + u(t-2) = -t^2 - 2t + 5, \tag{3.7.8}$$

this time with different initial function

$$\phi(t) = -t^2 - 4t - 1. \tag{3.7.11}$$

We show that using DTM combined with the method of steps we obtain different solution than in Example 19.

Remark first that $\phi'(t) = -2t - 4$. Applying substitutions results in

$$u''(t) - t\big[-2(t-1) - 4\big] + \big[-(t-2)^2 - 4(t-2) - 1\big] = -t^2 - 2t + 5.$$

Rearranging the terms, we obtain

$$u''(t) = -2t^2 - 4t + 2.$$

Applying DTM, we have recurrence relation

$$U(k+2) = \frac{-2\delta(k-2) - 4\delta(k-1) + 2\delta(k)}{(k+1)(k+2)}, \ k \geq 0 \tag{3.7.12}$$

in this case. From initial conditions $u(0) = 0$, $u'(0) = -4$, we have $U(0) = 0$, $U(1) = -4$ and from (3.7.12) we get $U(2) = 1$, $U(3) = -2/3$, $U(4) = -1/6$, $U(k) = 0$ for $k \geq 5$. Thus we have unique exact solution of Cauchy problem (3.7.8), (3.7.11) in the form

$$u(t) = \begin{cases} -t^2 - 4t - 1, \ t \in [-2, 0], \\ -1 - 4t + t^2 - \frac{2}{3}t^3 - \frac{1}{6}t^4, \ t \in [0, 1]. \end{cases}$$

However, it is not a solution on the whole interval $[0, \infty)$ in this case.

Example 21. Consider Cauchy problem consisting of first order differential equation with one constant delay

$$u'(t) = \frac{1}{a}u(t) - \frac{1}{a}u(t-a) + a, \ a > 0, \tag{3.7.13}$$

and initial function

$$\phi(t) = t^2, \ t \in [-a, 0]. \tag{3.7.14}$$

We can see that $\phi(t-a) = (t-a)^2 = t^2 - 2at + a^2$. Then using method of steps leads to equation

$$u'(t) = \frac{1}{a}u(t) - \frac{1}{a}\big[t^2 - 2at + a^2\big] + a.$$

Applying DTM, we derive relation

$$(k+1)U(k+1) = \frac{1}{a}U(k) - \frac{1}{a}\big[\delta(k-2) - 2a\delta(k-1) + a^2\delta(k)\big] + a\delta(k),$$

which can be simplified to

$$(k+1)U(k+1) = \frac{1}{a}U(k) - \frac{1}{a}\delta(k-2) + 2\delta(k-1).$$

From initial function (3.7.14) we deduce initial condition $u(0) = 0$, which is transformed by differential transformation to $U(0) = 0$. For $k = 0, 1, 2, \dots$ we have

$$\begin{aligned} U(1) &= \frac{1}{a}U(0) & &\to U(1) = 0, \\ 2U(2) &= \frac{1}{a}U(1) + 2 & &\to U(2) = 1, \\ 3U(3) &= \frac{1}{a}U(2) - \frac{1}{a} & &\to U(3) = 0, \\ &\vdots & &\vdots \end{aligned}$$

which gives unique solution of Cauchy problem (3.7.13), (3.7.14) on $t \in [0, a]$,

$$u(t) = t^2, \ t \in [0, a].$$

Obviously, $u(t) = t^2$ is unique solution of Cauchy problem (3.7.13), (3.7.14) on $[0, \infty)$, which can be easily verified.

Example 22. Consider delayed differential equation of the third order

$$u'''(t) = -u(t) - u(t - 0.3) + \mathrm{e}^{-t+0.3} \tag{3.7.15}$$

subject to the initial function

$$\phi(t) = \mathrm{e}^{-t}, \ t \le 0 \tag{3.7.16}$$

and conditions

$$\begin{aligned} u(0) &= 1, \\ u'(0) &= -1, \\ u''(0) &= 1. \end{aligned} \tag{3.7.17}$$

Straightforward observation gives us the information that it is enough to consider only (3.7.15) and (3.7.16), since conditions (3.7.17) are not independent of initial function $\phi(t)$ defined in (3.7.16). We mention conditions (3.7.17) to demonstrate that initial conditions are not sufficient to set Cauchy problem for DDE properly, we indeed need the whole initial function. We will see this in Example 23.

Let us start with the method of steps. Since $u(t-0.3) = \phi(t-0.3) = \mathrm{e}^{-(t-0.3)}$, we immediately see that equation (3.7.15) changes to ordinary differential equation

$$u'''(t) = -u(t).$$

Applying DTM gives simple recurrence relation

$$U(k+3) = \frac{-U(k)}{(k+1)(k+2)(k+3)}, \quad k \ge 0. \tag{3.7.18}$$

From initial conditions (3.7.17) and recurrence relation (3.7.18) we have

$$U(0) = 1, \ U(1) = -1, \ U(2) = \frac{1}{2}, \ U(3) = \frac{-1}{3!}, \ U(4) = \frac{1}{4!}, \dots, U(k) = \frac{(-1)^k}{k!}, \dots$$

Using inverse differential transformation (3.1.2) we obtain a solution of (3.7.15), (3.7.16) in the form

$$u(t) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} t^k = \mathrm{e}^{-t}. \tag{3.7.19}$$

It is the closed form unique solution of Cauchy problem (3.7.15), (3.7.16) on $[0, 0.3]$. It is not difficult to verify that it is a solution on the whole real axis (look at the initial function).

Example 23. Now consider equation

$$u'''(t) = -u(t) - u(t-0.3) + \mathrm{e}^{-t+0.3} \tag{3.7.15}$$

subject to initial function

$$\phi(t) = \frac{1}{2}t^2 - t + 1, \ t \in [-0.3, 0]. \tag{3.7.20}$$

This function satisfies conditions (3.7.17) as well. However, applying method of steps we obtain ordinary differential equation

$$u'''(t) = -u(t) - \frac{1}{2}t^2 - 1.3t + 1.39 + \mathrm{e}^{0.3}\mathrm{e}^{-t}.$$

Classical calculation (e.g. method of undetermined coefficients) shows that function

$$\begin{aligned} u(t) = &\left(2.23 + \frac{\mathrm{e}^{0.3}}{3}\right)\mathrm{e}^{-t} + \left(0.16 - \frac{\mathrm{e}^{0.3}}{3}\right)\mathrm{e}^{\frac{t}{2}}\cos\left(\frac{\sqrt{3}}{2}t\right) + \left(\frac{-0.3}{\sqrt{3}} + \frac{\mathrm{e}^{0.3}}{3\sqrt{3}}\right)\mathrm{e}^{\frac{t}{2}}\sin\left(\frac{\sqrt{3}}{2}t\right) \\ &+ \frac{\mathrm{e}^{0.3}}{3}t\mathrm{e}^{-t} - \frac{1}{2}t^2 + 1.3t - 1.39 \end{aligned} \tag{3.7.21}$$

is unique solution of Cauchy problem (3.7.15), (3.7.20) on interval $[0, 0.3]$, nevertheless it is completely different from (3.7.19). Using of DTM leads to recurrence relation

$$(k+1)(k+2)(k+3)U(k+3) = -U(k) - \frac{1}{2}\delta(k-2) + 1.3\ \delta(k-1) - 1.39\ \delta(k) + \mathrm{e}^{0.3}\frac{(-1)^k}{k!},$$

which together with conditions (3.7.17) transformed to $U(0) = 1$, $U(1) = -1$, $U(2) = \frac{1}{2}$ enables us to calculate

$$\begin{aligned} &U(3) = \frac{-2.39 + \mathrm{e}^{0.3}}{3!}, \ U(4) = \frac{2.3 - \mathrm{e}^{0.3}}{4!}, \quad U(5) = \frac{-1 + \mathrm{e}^{0.3}}{5!}, \\ &U(6) = \frac{2.39 - 2\mathrm{e}^{0.3}}{6!}, \ U(7) = \frac{-2.3 + 2\mathrm{e}^{0.3}}{7!}, \ U(8) = \frac{1 - 2\mathrm{e}^{0.3}}{8!}, \\ &U(9) = \frac{-2.39 + 3\mathrm{e}^{0.3}}{9!}, \quad \ldots \end{aligned}$$

Precise investigation of this sequence reveals that $U(k)$, $k = 0, 1, 2, \ldots$ are coefficients of Taylor expansion of function (3.7.21) in neighbourhood of 0, thus the calculation resulted in an approximation of unique solution of Cauchy problem (3.7.15), (3.7.20) on $[0, 0.3]$ as expected.

Example 24. Consider the following Cauchy problem for first order neutral differential equation

$$u'(t) + \frac{1}{4}u'(t-1) = u(t) + u(t-1), \ t \geq 0, \tag{3.7.22}$$

$$u(t) = -t, \quad \text{for } t \in [-1, 0]. \tag{3.7.23}$$

The exact solution which can be calculated classically by method of steps is given by

$$u(t) = t - \frac{1}{4} + \frac{1}{4}\mathrm{e}^t, \ t \in [0, 1]. \tag{3.7.24}$$

We remark that the so called *sewing condition*

$$\phi'(0) + \frac{1}{4}\phi'(-1) = \phi(0) + \phi(-1)$$

is not fulfilled hence the derivative of solution of problem (3.7.22), (3.7.23) is not continuous at 0.

Applying method of steps leads to equation

$$u'(t) - \frac{1}{4} = u(t) - t + 1.$$

Using DTM, we obtain relation

$$(k+1)U(k+1) - \frac{1}{4}\delta(k) = U(k) - \delta(k-1) + \delta(k),$$

which together with initial condition $u(0) = 0$ acquired from (3.7.23) and transformed to $U(0) = = 0$ implies

$$\begin{aligned}
U(1) - \frac{1}{4} &= U(0) + 1 & &\to U(1) = \frac{5}{4}, \\
2U(2) &= U(1) - 1 & &\to U(2) = \frac{1}{8} = \frac{1}{4 \cdot 2!}, \\
3U(3) &= U(2) & &\to U(3) = \frac{1}{24} = \frac{1}{4 \cdot 3!}, \\
4U(4) &= U(3) & &\to U(4) = \frac{1}{4 \cdot 4!}, \\
&\vdots & &\vdots
\end{aligned}$$

If we write $U(0)$ and $U(1)$ as $U(0) = \frac{1}{4 \cdot 0!} - \frac{1}{4}$ and $U1) = \frac{1}{4 \cdot 1!} + 1$ and perform inverse transformation, the solution can be expressed as

$$u(t) = \frac{1}{4 \cdot 0!} - \frac{1}{4} + \left(\frac{1}{4 \cdot 1!} + 1\right) t + \frac{1}{4 \cdot 2!} t^2 + \frac{1}{4 \cdot 3!} t^3 + \frac{1}{4 \cdot 4!} t^4 + \cdots + \frac{1}{4 \cdot k!} t^k + \cdots = t - \frac{1}{4} + \frac{1}{4} \sum_{k=0}^{\infty} \frac{1}{k!} t^k, \tag{3.7.25}$$

which is Taylor expansion of exact solution (3.7.24). Hence using the algorithm we are able to identify unique solution of Cauchy problem (3.7.22), (3.7.23) in closed form.

For the sake of completeness, we add some formulas which have been used in literature for solving of certain classes of equation (3.7.5).

Theorem 3.6 ([11]). *Assume that $F(k)$, $G(k)$ are differential transformations of functions $f(t)$, $g(t)$, where $a > 0$ is a real constant. If*

$$f(t) = g(t - a), \quad \text{then} \quad F(k) = \sum_{i=k}^{N} (-1)^{i-k} \binom{i}{k} a^{i-k} G(i), \ N \to \infty. \tag{3.7.26}$$

Proof. The proof follows immediately from definition of differential transformation and binomial formula:

Hence

$$\begin{aligned} u(t) &= \sum_{k=0}^{\infty} G(k)(t-t_0-a)^k = \sum_{k=0}^{\infty} G(k) \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} (t-t_0)^i a^{k-i} \\ &= \sum_{k=0}^{\infty} \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} (t-t_0)^i a^{k-i} G(k) = \sum_{i=0}^{\infty} \sum_{k=i}^{\infty} (t-t_0)^i (-1)^{k-i} \binom{k}{i} a^{k-i} G(k) \\ &= \sum_{i=0}^{\infty} (t-t_0)^i \sum_{k=i}^{\infty} (-1)^{k-i} \binom{k}{i} a^{k-i} G(k) = \sum_{k=0}^{\infty} (t-t_0)^k \sum_{i=k}^{\infty} (-1)^{i-k} \binom{i}{k} a^{i-k} G(i). \end{aligned}$$

The proof is complete. □

Using Theorem 3.6 and the formula i) in Theorem 3.1, differential transformation formula for function $f(t) = \frac{d^n}{dt^n} g(t-a)$ can be deduced.

Theorem 3.7 ([11]). *Assume that $F(k)$, $G(k)$ are differential transformations of functions $f(t)$, $g(t)$, $a > 0$. If*

$$f(t) = \frac{d^n}{dt^n} g(t-a), \quad \text{then} \quad F(k) = \frac{(k+n)!}{k!} \sum_{i=k+n}^{N} (-1)^{i-k-n} \binom{i}{k+n} a^{i-k-n} G(i), \ N \to \infty. \tag{3.7.27}$$

Using Theorems 3.6, 3.7 and formula vii) in Theorem 3.1 any differential transformation of a product of functions with delayed arguments and derivatives of that functions can be proved. However, such formulas are complicated and not generally applicable for solving functional differential equations with multiple constant delays (see for example [11], [12], [13]).

4 Exercises

4.1 Exercises

Example 1: Use DTM to solve the following initial value problems for ordinary differential equations:

a) $u' = u^2, \quad u(0) = 1.$

b) $u' = u - u^2, \quad u(0) = 2.$

c) $u'' + (1-t)u' - u - 2u^3 = 0, \quad u(0) = u'(0) = 1.$

d) $u'' + u' + u^3 = 0, \quad u(0) = u'(0) = 1.$

e) $u^{(4)} + uu'' - u' = 0, \quad u(0) = 0,\ u'(0) = u''(0) = u'''(0) = 1.$

Example 2: Use DTM to solve the following initial value problems for systems of ordinary differential equations:

a)

$$\begin{aligned} u_1' &= 2u_1 - u_2 + 2, \quad u_1(0) = 0 \\ u_2' &= 4u_1 - 2u_2 + 2, \quad u_2(0) = 1. \end{aligned}$$

b)

$$\begin{aligned} u_1' &= 2u_1 - 3u_2, \quad u_1(0) = 8 \\ u_2' &= -2u_1 + u_2 + 2, \quad u_2(0) = 3. \end{aligned}$$

c)

$$\begin{aligned} u_1' &= -u_2' - u_1 - u_2 + 1, \quad u_1(0) = 0 \\ u_2' &= 2u_1 + u_2, \quad u_2(0) = 1. \end{aligned}$$

d)

$$\begin{aligned} u_1'^2 &= -u_2 + 1, \quad u_1(0) = u_1'(0) = 0 \\ u_2'^2 &= -u_1, \quad u_2(0) = u_2'(0) = 0. \end{aligned}$$

Example 3: Use DTM to solve the following singular initial value problems:

a) $u'' + \frac{2}{t}u' = (4x^2 - 6)u, \quad u(0) = 1,\ u'(0) = 0.$

b) $u'' + \frac{2}{t}u' + u = t^5 + 30t^3, \quad u(0) = u'(0) = 0.$

c) $u'' + \frac{8}{t^2}u' + 18u + 4u\ln u = 0, \quad u(0) = 1,\ u'(0) = 0.$

d) $u'' + \frac{2}{t}u' + u^3 = 6 + t^3, \quad u(0) = u'(0) = 0.$

e) $u'' + \frac{8}{t}u' + tu = t^5 - t^4 + 44t^2 - 30t, \quad u(0) = u'(0) = 0.$

Example 4: Use the general formula for DTM to solve the following initial value problems:

a) $u'' = 2u + 4u\ln u, \quad u(0) = 1,\ u'(0) = 0.$

b) $u'' = \frac{8u^2}{1+2x}, \quad u(0) = 1,\ u'(0) = 2.$

c) $u' = -\frac{1}{\sin u}, \quad u(0) = \pi/2.$

c) $u' = \frac{u^2}{1+u}, \quad u(0) = 1.$

Example 5: Use DTM to solve the following initial value problems for differential equations with proportional delays:

a) $u'' = \frac{3}{4}u + u(\frac{t}{2}) - t^2 + 2, \quad u(0) = 0,\ u'(0) = 0.$

b) $u''' = u + u'(\frac{t}{2}) + u''(\frac{t}{3}) + \frac{1}{2}u'''(\frac{t}{4}), \quad u(0) = u'(0) = u''(0) = 0.$

c) $u'' = u'(\frac{t}{2}) + \frac{1}{2}u''(\frac{t}{2}) + 2, \quad u(0) = 1,\ u'(0) = 0.$

d)

$$\begin{aligned} u_1'(t) &= u_2^2(t), \quad u_1(0) = 1 \\ u_2'(t) &= \frac{1}{2}u_1(\frac{t}{2}), \quad u_2(0) = 1 \\ u_3'(t) &= \mathrm{e}^{5/2t}u_2(t) + 9\mathrm{e}^{2t}u_3(\frac{t}{3}), \quad u_3(0) = 0. \end{aligned}$$

Example 6: Use combination of method of steps and DTM to solve the following Cauchy problems for differential equations with constant delays:

a) $u'(t) + u(t-1) + u(t-2) = 2t - 2$ for $t \in [0,1]$, initial function $\phi(t) = 0$ on $[-2,0]$.

b) $u'(t) + u(t-1) + u(t-2) = 2t - 2$ for $t \in [0,1]$, initial function $\phi(t) = \mathrm{e}^t$ on $[-2,0]$.

c) $u'(t) - u(t-\frac{\pi}{2}) = \sin t$ for $t \in [0,1]$, initial function $\phi(t) = t$ on $[-\frac{\pi}{2},0]$.

d)

$$\begin{aligned} &u_1'(t) + u_1(t)u_2(t-1) = \mathrm{e}^t && \text{for } t \in [0,1], \quad \text{initial function } \phi_1(t) = t^2 \text{ on } [-1,0], \\ &u_2'(t) + u_1(t-1)u_2(t) = \mathrm{e}^{-t} && \text{for } t \in [0,1], \quad \text{initial function } \phi_2(t) = t \text{ on } [-1,0]. \end{aligned}$$

4.2 Results

Example 1

a) For $n = 4$:

$$u(t) = 1 + t + t^2 + t^3 + \cdots \Rightarrow \text{ for } n \to \infty, \quad u(t) = \frac{1}{1-t}.$$

b) For $n = 5$:

$$u(t) = 2 - 2t + 3t^2 - \frac{13}{3}t^3 + \frac{25}{4}t^4 + \cdots \Rightarrow \text{ for } n \to \infty, \quad u(t) = \frac{2}{2 - \mathrm{e}^{-t}}.$$

c) For $n = 4$:

$$u(t) = 1 + t + t^2 + t^3 + \cdots \Rightarrow \text{ for } n \to \infty, \quad u(t) = \frac{1}{1-t}.$$

d) For $n = 4$ we obtain only approximate solution:

$$u(t) = 1 + t - t^2 - 0,166666t^3 + 0,041667t^4 + \ldots.$$

e) For $n = 4$:

$$u(t) = t + \frac{t^2}{2!} + \frac{t^3}{3!} + \frac{t^4}{4!} + \cdots \Rightarrow \text{ for } n \to \infty, \quad u(t) = \mathrm{e}^t - 1.$$

Example 2.

a)

$$\begin{aligned} u_1 &= t^2 + t, \\ u_2 &= 2t^2 - t + 1. \end{aligned}$$

b) For $n = 4$:

$$\begin{aligned} u_1(t) &= 8 + 7t + \frac{53}{2}t^2 + \frac{187}{6}t^3 + \cdots \Rightarrow \text{for } n \to \infty, \quad u_1(t) = 5\mathrm{e}^{-t} + 3\mathrm{e}^{4t}, \\ u_2(t) &= 3 - 13t - \frac{27}{2}t^2 - \frac{135}{6}t^3 - \cdots \Rightarrow \text{for } n \to \infty, \quad u_2(t) = 5\mathrm{e}^{-t} - 2\mathrm{e}^{4t}. \end{aligned}$$

c) For $n = 5$:

$$\begin{aligned} u_1(t) &= -t + t^2 - \frac{5}{6}t^3 + \frac{5}{8}t^4 - \frac{7}{120}t^5 - \cdots \Rightarrow \text{for } n \to \infty, \quad u_1(t) = \mathrm{e}^{-t} - 1, \\ u_2(t) &= 1 + t - \frac{1}{2}t^2 + \frac{1}{2}t^3 - \frac{7}{24}t^4 + \cdots \Rightarrow \text{for } n \to \infty, \quad u_2(t) = 2 - \mathrm{e}^{-t}. \end{aligned}$$

d) For $n = 4$:

$$u_1(t) = \frac{t^2}{2} - \frac{t^3}{6} + \frac{t^4}{12} + \frac{t^5}{5} + \cdots \Rightarrow \text{for } n \to \infty, \quad u_1(t) = 1/4(\mathrm{e}^t + \mathrm{e}^{-t} - 2\cos t),$$

$$\begin{aligned} u_2(t) &= -\frac{t^4}{24} - \frac{t^5}{120} - \frac{t^6}{360} - \frac{t^7}{840} + \cdots \Rightarrow \text{for } n \to \infty, \\ u_2(t) &= 1/4(4 - \mathrm{e}^t - \mathrm{e}^{-t} - 2\cos t). \end{aligned}$$

Example 3.

a) For $n = 4$:

$$u(t) = 1 + t^2 + \frac{t^4}{2} + \frac{1}{6}t^6 + \cdots \Rightarrow \text{ for } n \to \infty, \quad u(t) = \mathrm{e}^{t^2}.$$

b) $u(t) = t^5$.

c) For $n = 4$:

$$u(t) = 1 - t^2 + \frac{t^4}{2} - \frac{1}{6}t^6 + \cdots \Rightarrow \text{ for } n \to \infty, \quad u(t) = \mathrm{e}^{-t^2}.$$

d) $u(t) = t^2$.

e) $u(t) = t^4 - t^3$.

Example 4.

a) For $n = 4$:

$$u(t) = 1 + t^2 + \frac{t^4}{2} + \frac{1}{6}t^6 + \cdots \Rightarrow \text{ for } n \to \infty, \quad u(t) = \mathrm{e}^{t^2}.$$

b) For $n = 4$:

$$u(t) = 1 - 2t + 4t^2 - 8t^3 + 16t^4 - 32t^5 + \cdots \Rightarrow \text{ for } n \to \infty, \quad u(t) = \frac{1}{1-2t}.$$

c) For $n = 4$:

$$u(t) = -\frac{\pi}{2} - t - \frac{1}{6}t^3 - \frac{3}{40}t^5 + \cdots \Rightarrow \text{ for } n \to \infty, \quad u(t) = \arccos t.$$

d) For $n = 4$:

$$u(t) = 1 + \frac{1}{2}t + \frac{3}{16}t^2 + \frac{7}{96}t^3 + \ldots$$

Example 5.

a) $u(t) = t^2$.

b) $u(t) = t^4$.

c) $u(t) = 1 + t^2$.

d) $u_1(t) = \mathrm{e}^t,\ u_2(t) = \mathrm{e}^{t/2},\ u_3(t) = t\mathrm{e}^{3t}$.

Example 6.

a) $u(t) = t^2 - 2t$.

b) For $n = 4$:

$$u(t) = 1 + \left(1 - \frac{1}{\mathrm{e}} - \frac{1}{\mathrm{e}^2}\right)\left(t + \frac{t^2}{2} + \frac{t^3}{3!} + \frac{t^4}{4!} + \ldots\right) \Rightarrow \text{ for } n \to \infty, \quad u(t) = \frac{1}{\mathrm{e}} + \frac{1}{\mathrm{e}^2} + \left(1 - \frac{1}{\mathrm{e}} - \frac{1}{\mathrm{e}^2}\right)\mathrm{e}^t.$$

c) For $n = 8$:

$$u(t) = -\frac{\pi}{2}t + t^2 - \frac{t^4}{4!} + \frac{t^6}{6!} - \frac{t^8}{8!} + \cdots \Rightarrow \text{ for } n \to \infty, \quad u(t) = -\frac{\pi}{2}t + \frac{t^2}{2} + 1 - \cos t.$$

d) For $n = 4$:

$$\begin{aligned} u_1(t) &= t + t^2 + \frac{1}{3!}t^3 + \frac{5}{4!}t^4 + \ldots, \\ u_2(t) &= t - t^2 + \frac{7}{3!}t^3 - \frac{26}{4!}t^4 + \ldots. \end{aligned}$$

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